THE ECHO PERIOD OF CHOPTUIK'S CRITICAL SOLUTION AT LARGE DIMENSION: A SELECTION RULE AND A COMPUTER-ASSISTED PROOF

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Abstract. Choptuik's critical solution of spherical scalar-field collapse is discretely self-similar, with an echo period Δ that depends on the spacetime dimension $D = n + 2$; computed to $D = 61$, it approaches 1.643893746 as $D \to \infty$, if it is analytic in $1/D$. We study the limit as a selection problem of exponential asymptotics. Within a scaling ansatz measured on the solutions, the $1/n$ expansion of the self-similar field equations is unobstructed to all orders for any period and any time profile, so the period is selected beyond all orders. At leading order the problem is solved by ${}_2F_1$, with a closed-form, never-vanishing light-cone coefficient; gravity transports the corresponding fast WKB mode by a non-vanishing factor; and the late orders of the $1/n$ series encode the coefficient as the Stokes multiplier of a mode born at infinity. At $x \sim \sqrt{n}$ the equations reduce to an n-free inner problem, the leading-order large-D system of Rozali and Way, and we conjecture that the period is selected by the vanishing of its fast-type Stokes multipliers. In the inner problem a shift of r^2 gives an exact unstable mode with growth rate 2, which yields the critical exponent 1/2 under the conjecture and two further hypotheses. We reduce a selection theorem for models with finitely many harmonics to lemmas, certify its two contraction arguments for one harmonic, and prove by a computer-assisted argument, assuming Braaksma's summability theorem, that the selection condition of the one-harmonic model has exactly one zero in an explicit box, and that it is simple. Evaluated numerically without truncation, the conjectured rule gives $\Delta_\infty = 1.6439 \pm 0.0001$ if the selection function has no other nearby zeros. The work was done by an AI model (Claude, Anthropic) under the author's direction and has not been reviewed by a human expert.

1. Introduction

Choptuik's critical solution [13] is the spacetime at the threshold of black-hole formation in the spherically symmetric collapse of a massless scalar field. It is discretely self-similar (DSS): in suitable coordinates it repeats itself on scales smaller by a fixed factor e^Δ, and Δ is called the echo period. As a mathematical object it is the solution of a nonlinear eigenvalue problem [24]: a solution of the self-similar Einstein–scalar equations that is periodic in logarithmic time, regular at the centre and analytic at the past light cone of the singularity exists only for isolated values of Δ. In four spacetime dimensions $\Delta = 3.445452402(3)$ [33], and Reiterer and Trubowitz proved the existence of the solution by a computer-assisted argument that encloses Δ to about 80 digits [43]. The problem can be posed in any number $D = n + 2$ of spacetime dimensions, with n entering the equations as a parameter; see the review [25] for the physics.

This paper is about the limit $n \to \infty$. In the companion paper [1] the critical solution is computed up to $D = 61$, and the echo period approaches a finite limit, $\Delta_\infty = 1.643893746$ if Δ is analytic in $1/D$ (Section 2.2). The questions here are mathematical. Why does the expansion of the equations in powers of $1/n$ not determine Δ_∞? What does? And what can be proved about it?

The answer has the structure of a classical selection problem of exponential asymptotics. In Saffman–Taylor finger selection and in dendritic growth, a formal expansion in a small parameter exists for every value of the quantity to be selected, and an actual solution exists only when an exponentially small term vanishes; that term is the Stokes multiplier of an inner equation posed near a complex singular point [15, 26, 29]. Xie and Tanveer made this rigorous [49, 46, 48]. In our problem the small parameter is $1/n$, the selected quantities are the frequency $\omega = 2\pi/\Delta$ and the time profile of the solution, the symmetry condition is analyticity at the light cone, and the complex singular point is the point at infinity of the self-similar radius. Here the unknown depends on the radius and a periodic phase, and the inner problem is an infinite system coupled through harmonics.

Results. We work within a scaling ansatz for large n that is measured on the numerical solutions (Ansatz 2.2).

(1) The formal $1/n$ expansion exists to all orders for every period and every time profile (Proposition 1), so the period is selected beyond all orders in $1/n$.

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TABLE 1. The statements of this paper and their status.

Statement	Content	Status
Proposition 1	$1/n$ expansion unobstructed to all orders	proved (within Ansatz 2.2)
Proposition 2	leading order by ${}_2F_1$; closed-form $C(n,s) \neq 0$	proved
Proposition 3	transport of the fast WKB mode with gravity	proved (to the order stated)
Proposition 4	late orders of the leading-order series; $C(n,s)$ as a Stokes multiplier	proved
Conjecture 1	selection at $D = \infty$ by the inner multipliers	conjecture, with argument
Proposition 5	exact $\lambda = 2$ mode of the inner problem	proved; part (i) checked symbolically
Corollary 1	$\gamma_\infty = 1/2$	conditional on Conjecture 1 and two hypotheses
Theorem B(M)	selection for the M-harmonic model	conditional on the lemmas of Table 2; for $M = 1$ not yet proved (Section 6.5)
Lemmas B1, B5.1, B6′, B8.1, B8.2; B2, B5.0	ingredients of Theorem B(M)	proved; B2 and B5.0 checked symbolically
Theorem A($M = 1$)	one simple zero of the $M = 1$ selection condition	certified (computer-assisted), assuming Braaksma's theorem
Section 7.11	one simple zero of the $M = 3$ selection condition	certified (computer-assisted), assuming Braaksma's theorem; not independently reviewed
Lemma 8.1	enclosure of the first amplitude coefficient σ_1	certified (computer-assisted, `exact_sigma.py`); not independently reviewed
Section 9	$\omega^*/\omega_\infty = 0.99992$–$1.00005$ at fourteen orders (no other zeros of S_1 near x_0 assumed)	numerical evidence

(2) The leading order is a free wave solved by ${}_2F_1$, with an explicit, never-vanishing light-cone coefficient $C(n,s)$ (Proposition 2); gravity transports the corresponding fast WKB mode by a non-vanishing factor (Proposition 3); and the late orders of the $1/n$ series encode $C(n,s)$ as the Stokes multiplier of an exponential born at $x = \infty$ (Proposition 4).

(3) At $x \sim \sqrt{n}$ lives an n-free inner problem, which is the leading-order large-D system of Rozali and Way [44] in other variables. Without gravity its Stokes constant is Weber's and reproduces $C(n,s)$. We conjecture that the $D = \infty$ solution is selected by the vanishing of its fast-type Stokes multipliers (Conjecture 1), and give the argument, its hypotheses and the three places where rigour fails.

(4) In $s = r^2/(2n)$ the inner problem is an autonomous nonlinear heat equation, whose s-shift, gauge-fixed, is an exact perturbation with growth rate 2 (Proposition 5); under Conjecture 1, uniqueness of the unstable mode and continuity in $1/n$, the critical exponent tends to $1/2$ (Corollary 1).

(5) A selection theorem for M-harmonic models (Theorem B(M)), after Xie and Tanveer, is reduced to lemmas. The two contraction arguments among them, the global continuation and the inner–outer matching, are proved for each fixed M and carried out with computer-assisted constants for $M = 1$ and $n \geq e^{73.89}$; what the one-harmonic theorem still lacks is listed in Section 6.5.

(6) For the one-harmonic truncation, the selection condition has exactly one zero in an explicit box of half-width 10^{-9}, and it is simple (Theorem A($M = 1$)). The proof is computer-assisted, assumes only Braaksma's summability theorem, and follows Baldomá, Capiński, Guardia and Seara [4].

(7) The untruncated multiplier is numerically an entire function of the squared amplitude, of exponential type; its first coefficient is enclosed rigorously, by a computer-assisted computation not yet independently reviewed (Section 8).

Evaluated numerically to twelve orders, the selection rule gives $\Delta_\infty \approx 1.644$ from the $D = \infty$ equations alone: evidence for Conjecture 1, with the fourth digit fixed only under an assumption on the selection function's other zeros (Section 9). Table 1 lists the statements and their status.

Prior work. The leading-order freedom of the period, any periodic profile with any period, was found by Ecker, Ecker and Grumiller [22], who constructed analytic DSS solutions at large D in the same gauge; Proposition 1 extends it to all orders (Remark 3.1 discusses their next-to-leading-order condition). The free wave equation in similarity coordinates was analysed in $3 + 1$ dimensions by Donninger [21], which contains the $D = 4$ case of Proposition 2; the coefficient $C(n,s)$ follows from Gauss's connection formula [20, Eq. 15.8.4] once the mode equation is recognised as hypergeometric, and we did not find it for general D in the literature we

searched. Proposition 4 is an explicit instance of Dingle's late-order analysis [19], whose nonlinear form is due to Chapman, King and Adams [12]; the analogous singularity structure for ${}_2F_1$ with large parameters is worked out by Paris [40]. Trans-series corrections in $1/n$ are familiar in large-D gravity [23]. Rozali and Way [44] derived the D-free leading-order equations that are our inner problem, noted that they are parabolic with the roles of space and time swapped, and estimated $\gamma = 1/2$ from the divergence of their oscillating solutions. What is new here is their use for the discretely self-similar solution: the inner Stokes multipliers, the selection conjecture and the symmetry argument of Proposition 5. Clark and Pimentel [14] constructed continuously self-similar critical solutions at large D by matched asymptotics. The proof templates are Xie and Tanveer (selection), Baldomá, Fontich, Guardia and Seara [5] (matching and continuation) and Baldomá, Capiński, Guardia and Seara [4] (the computer-assisted Stokes constant). Other rigorous instances of "existence if and only if an exponentially small coefficient vanishes" are in Lombardi's monograph [32].

Organization. Section 2 states the problem, the numerical input and the scaling ansatz. Section 3 proves Propositions 1–4. Section 4 introduces the inner problem, its Stokes structure and Conjecture 1, and Section 5 proves Proposition 5. Section 6 sets out the selection theorem for M-harmonic models, Section 7 the computer-assisted proof of Theorem A($M = 1$), and Section 8 the entire-function structure of the untruncated multiplier. Section 9 collects the numerical evidence for the conjecture, and Section 10 the open problems. Every computation quoted here is reproducible from the repository [2]; script names refer to it.

2. Setting

2.1. **The self-similar equations.** Let $D = n + 2$ be the spacetime dimension and $\varepsilon = 1/n$. We use the self-similar radius x (areal radius over time to the singularity, normalized so that the past light cone of the singularity is at $x = 1$), logarithmic time $\tau = -\ln(-t)$, and the phase $\theta = \omega\tau$, with $\omega = 2\pi/\Delta$. The unknowns are four functions of (x, θ), 2π-periodic in θ: p and m, the scaled derivatives of the scalar field along the two radial null directions; $b = a^2$, with a the radial metric function; and c, the lapse-type function that sets the light cones (the past light cone is where $x = c$). With $G = (n-1)\,c\,(b-1)$, the self-similar Einstein–scalar equations read [1]

$$xc_x = (n-1)\,c\,(b-1), \qquad xb_x = b\,(p^2+m^2) - (n-1)\,b\,(b-1), \tag{2.1}$$

$$x(x-c)\,p_x = -\omega x\,p_\theta + Gp - cp + \tfrac{n}{2}\,c\,(p+m), \tag{2.2}$$

$$x(x+c)\,m_x = -\omega x\,m_\theta - Gm + cm - \tfrac{n}{2}\,c\,(p+m). \tag{2.3}$$

Definition 2.1 (the DSS problem)**.** *Find $\omega > 0$ and functions p, m, b, c on $0 \le x \le 1$, 2π-periodic in θ, with only odd harmonics of θ in p, m and only even ones in b, c, that solve (2.1)–(2.3) and are*

(i) regular at the centre $x = 0$,
(ii) analytic at the past light cone $x = 1$, and
(iii) normalized by $c(1) = 1$ (the gauge that puts the light cone at $x = 1$).

Both $x = 0$ and $x = c = 1$ are regular singular points of (2.2)–(2.3). Regularity at the centre removes one branch at $x = 0$; at $x = 1$ the general solution has a branch $(1-x)^\Lambda$ with non-integer Λ, and requiring its coefficients to vanish is what selects ω, as in [24]. The centre series has the exact symmetry $m(x) = p(-x)$.

2.2. **Numerical input.** The companion paper [1] computes the critical solution at 857 values $8.25 \le D \le 61.03$ in double-double arithmetic [18]. If Δ is analytic in $1/D$, which held-out tests there support but do not prove, $\Delta_\infty = 1.643893746$ and $2\pi/\Delta_\infty = 3.82213590$, to about 10^{-8} (an estimator range rather than a calibrated error; its construction is described there). The growth rate λ of the unstable mode, $\lambda = 1/\gamma$ with γ the critical exponent of the black-hole radius, $r_{\rm BH} \propto (p - p^*)^\gamma$ [27], tends to 2: free fits in $1/n$ put $\lambda_\infty = 2$ to about 10^{-6}. The time profile at the centre, $F_n(\theta) = \sqrt{n}\,\partial_x p(0,\theta)$, converges as D grows: its first harmonic extrapolates to about 2.72, and $n\,(1 - \langle c(0)\rangle)$ and $\langle F^2\rangle$ approach a common limit near 3.7 (angle brackets denote the mean over θ); the measured values are tabulated in [1]. None of these numbers is used in a proof below. Where an exact value of ω_∞ is needed, as in Theorem A($M = 1$), we *define* $\omega_\infty := 2\pi/1.64389375$, the reference value used in all the codes; this is a convention (it differs from $2\pi/\Delta_\infty$ by 2.4×10^{-9} relative), and the theorem is equally a statement about ω itself.

2.3. **The scaling at large n.** On the double-double solutions [1], $p \approx xF(\theta)/\sqrt{n}$ near the centre, $p+m$ is smaller than p by a further factor of order $1/n$, $c = 1 - O(1/n)$ and $b - 1 \approx (p^2+m^2)/(n-1)$. Write $p = \varepsilon^{1/2} P$, $m = \varepsilon^{1/2} M$, $S = P + M$ and $\Delta_P = P - M$. The sum and the difference of (2.2) and (2.3) are

$$x\,c\,\partial_x \Delta_P - c\,\Delta_P = x^2 \partial_x S + \omega x\,\partial_\theta S - \tfrac{1-\varepsilon}{\varepsilon}\,c\,(b-1)\,\Delta_P, \tag{2.4}$$

$$S = \varepsilon\,\frac{x^2 \partial_x \Delta_P - xc\,\partial_x S + \omega x\,\partial_\theta \Delta_P}{c\,(1-\varepsilon)\,b}. \tag{2.5}$$

Ansatz 2.2 (scaling)**.** The fields have formal expansions in integer powers of ε,

$$S = \sum_{k\geq 1} \varepsilon^k S_k, \quad \Delta_P = \sum_{k\geq 0} \varepsilon^k D_k, \quad c = 1 - \sum_{k\geq 1} \varepsilon^k G_k, \quad b = 1 + \sum_{k\geq 2} \varepsilon^k B_k, \quad \omega = \sum_{k\geq 0} \varepsilon^k \omega_k, \tag{2.6}$$

with coefficients that are functions of (x, θ), 2π-periodic in θ.

The scaling is forced by one measured fact, $c - 1 = O(1/n)$ (gravity is weak: $n(1 - \langle c(0)\rangle)$ converges). The b equation gives $b-1 \approx (p^2+m^2)/(n-1)$; the c equation then gives $c-1 = O(n(b-1)) = O(p^2)$, so $c-1 = O(1/n)$ requires $p = O(n^{-1/2})$; and the terms of order n in (2.2)–(2.3) force $p + m = O(p/n)$, which is the measured $p+m = O(n^{-3/2})$. Integer powers of ε beyond the overall $\varepsilon^{1/2}$ are an assumption (analyticity in $1/n$), consistent with the held-out tests on $\Delta(D)$ in [1] but not implied by them. All statements of Section 3 about the formal expansion hold within this ansatz; whether the formal family is the asymptotic expansion of actual solutions is hypothesis (H2) of Conjecture 1.

3. The outer expansion and its late orders

3.1. The $1/n$ expansion is unobstructed.

Proposition 1. *Let $F(\theta)$ be any smooth 2π-periodic function with odd harmonics only, $\omega_0 \neq 0$, $\omega_1, \omega_2, \ldots$ any numbers, and $K_1(\theta), K_2(\theta), \ldots$ any smooth odd-harmonic functions. Then (2.4)–(2.5) together with (2.1) have exactly one formal solution of the form (2.6) with $D_0 = 2xF$ in which, at every order,*

- *S_k is an even polynomial in x starting at x^2,*
- *$D_k - xK_k$ is an odd polynomial starting at x^3, and*
- *G_k and B_{k+1} are even polynomials (B starting at x^2) with $G_k(1) = 0$,*

all with smooth periodic coefficients. Every order is regular at the centre and analytic at the light cone, and $m(x) = p(-x)$ holds order by order. (The hypothesis $\omega_0 \neq 0$ only excludes the static case; the proof does not use it.)

Proof. By induction on k, with the structure in the statement as the hypothesis for all orders below k. Write E for the even polynomials in x with smooth periodic coefficients.

(1) B_{k+1}. The b equation at order $k+1$ gives B_{k+1} algebraically, since its coefficient is $(n-1)b \approx 1/\varepsilon$, from $x\partial_x B_k$, $(1+B)(P^2+M^2)$ and $(1-\varepsilon)(1+B)B$ at lower orders. Since $P^2 + M^2 = (S^2 + \Delta_P^2)/2$ with $S \in x^2 E$ and $\Delta_P \in xE$, every term lies in $x^2 E$.

(2) G_k. The c equation at order $k+1$ reads $-x\,\partial_x G_k = [(1-\varepsilon)\,c\,(b-1)]_{k+1} \in x^2 E$. Dividing by x leaves only odd powers x^{2j-1}, $j \geq 1$, so $G_k = -\sum_j r_{2j} x^{2j}/(2j) + \text{const}$ is an even polynomial without $\log x$. The constant is fixed by $G_k(1) = 0$.

(3) S_k. The numerator $x^2 \partial_x \Delta_P - xc\,\partial_x S + \omega x\,\partial_\theta \Delta_P$ of (2.5) maps $\Delta_P \in xE$, $S \in x^2 E$ into $x^2 E$. The denominator is $1 + O(\varepsilon)$ with coefficients in E, so its inverse is a power series in ε with coefficients in E. Hence $S_k \in x^2 E$.

(4) D_k. Equation (2.4) at order k is $x\,\partial_x D_k - D_k = R_k$. Here R_k collects $x^2 \partial_x S_k + \omega x\,\partial_\theta S_k \in x^3 E$, terms $c\,(b-1)\,\Delta_P \in x^3 E$, and terms $G_i\,(x\,\partial_x D_j - D_j)$; the last are in $x^3 E$ because $x\partial_x - 1$ annihilates xK_j. So R_k is odd and starts at x^3, and

$$D_k = x \int R_k\, x^{-2}\, \mathrm{d}x = \sum_{j\geq 1} \frac{R_{k,2j+1}}{2j}\, x^{2j+1} + xK_k.$$

The only possible obstruction, a term x^1 in R_k that would produce $x \log x$ (not allowed at the centre), never occurs.

(5) *Symmetry.* S even and Δ_P odd in x is the same as $p(-x) = m(x)$.

(6) *Harmonic content.* If P, M (hence S, Δ_P) contain only odd harmonics of θ and B, G only even ones at the lower orders, every step above preserves this: products of two odd-harmonic factors are even-harmonic, products of an odd and an even one are odd-harmonic, and ∂_θ keeps the harmonic.

Polynomials are analytic everywhere, so the light-cone condition holds automatically, and $c(1) = 1$ holds by construction. Nothing in the construction constrains F, ω_0 or the ω_k. □

The leading-order part of this freedom, any periodic profile with any period, is in [22].

Remark 3.1 (a published next-to-leading-order condition). Ecker, Ecker and Grumiller, in the same gauge (their $f = \alpha/a$ is our c), require $f \le 1$ near the light cone and obtain from it, at next-to-leading order, a condition at the zeros of the profile ($\omega^2 F'' + 3\omega F' = 0$ there, in our variables), which leads to their Eq. (20) and which they take to fix the period [22]; Lai and Wen use the same condition to select the period [30]. It does not constrain the formal solution of Proposition 1. The inequality holds for every regular solution, whatever the period: by (2.1) with $b(0) = 1$, b cannot fall below 1, so $xc_x \ge 0$ and $c \le c(1) = 1$. Near a zero of F the next-to-leading-order term of $b - 1$ is the cross term of a square whose square term arrives one order later, and the computed solutions do not satisfy the condition [1].

The first orders (computed symbolically, to fourth order with `solver/largeD_expand.py 4`) are

$$P_0 = xF, \qquad B_2 = 2x^2F^2, \qquad G_1 = F^2\,(1-x^2), \qquad S_1 = 2x^2\,(F + \omega_0 F'),$$
$$D_1 = x^3\,(\omega_0^2 F'' + 3\omega_0 F' + 2F - 2F^3) + xK_1. \tag{3.1}$$

Substituting the expansion through S_3, G_3, B_4 into the exact equations, with F from the $D = 60$ solution, the relative residual of every equation falls like n^{-3} for $n = 50, \dots, 800$, both for $\omega_0 = 2\pi/1.64389$ and for $\omega_0 = 2\pi/1.4$ (`solver/largeD_formal_check.py`).

Remark 3.2 (selection beyond all orders). If $\Delta(D)$ is analytic in $1/D$, its expansion coefficients cannot be computed order by order from the field equations: within Ansatz 2.2 the selection acts through terms smaller than every power of $1/n$. The same holds for the critical exponent. The expansion of growing perturbations $\propto e^{\lambda\tau}$ of the formal solution goes through for every λ, since the induction above works with every τ-derivative of a perturbation replaced by $\lambda + \omega\partial_\theta$ (checked symbolically through order 2, `solver/inner_pert/expand_pert.py`). So $\lambda = 1/\gamma$ is also selected beyond all orders.

3.2. **The leading-order problem.** At leading order ($c = b = 1$) the equations are the scalar wave equation on flat space in self-similar form. (In four dimensions Price and Pullin approximated the critical solution, in each of two regions, by flat-space scalar-field solutions [41].) For one harmonic $e^{\mathrm{i}\mu\theta}$ (μ odd) put

$$s = \mathrm{i}\mu\omega, \qquad \alpha = \tfrac{1+s}{2}, \qquad \beta = \tfrac{2+s}{2}, \qquad \kappa = \tfrac{n+1}{2}. \tag{3.2}$$

Proposition 2. *The solution of the leading-order problem that is regular at the centre is, up to a constant,*

$$p - m = x\,{}_2F_1(\alpha, \beta; \kappa; x^2), \qquad p + m = \frac{x(1-x^2)\,\partial_x(p-m) - (1+sx^2)\,(p-m)}{x\,(s-n+1)}. \tag{3.3}$$

Near the light cone it contains the non-analytic branch $(1-x^2)^{n/2-1-s}$ with coefficient

$$C(n,s) = \frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(s+1-\frac{n}{2}\right)}{\Gamma\left(\frac{1+s}{2}\right)\Gamma\left(\frac{2+s}{2}\right)}, \tag{3.4}$$

which is finite and non-zero for every real $\omega \neq 0$, odd μ and $n \ge 2$. Hence no time-periodic self-similar solution of the linear problem is analytic at the light cone, in any dimension $D \ge 4$.

Proof. With $c = b = 1$ the difference equation gives $(n-1)\,(p+m) = x^2\,\partial_x(p-m) - x\,\partial_x(p+m) + sx\,(p-m)$, and the sum gives $x\,\partial_x(p-m) - (p-m) = x^2\,\partial_x(p+m) + sx\,(p+m)$. Eliminating $p+m$ gives, for $y = p - m$,

$$x^2(1-x^2)\,y'' + x\left(n - 2 - 2(s+1)x^2\right)y' - \left(n - 2 + s(s+1)x^2\right)y = 0,$$

and with $y = x\,w(z)$, $z = x^2$, this is Gauss's equation $z(1-z)w'' + [\kappa - (\alpha+\beta+1)z]\,w' - \alpha\beta\,w = 0$. Its exponents are 0 and $1-\kappa$ at $z = 0$, so $y \sim x$ or $y \sim x^{2-n}$; for $n > 2$ regularity, and for $n = 2$ parity ($y = p - m$ is odd in x because $m(x) = p(-x)$, while x^0 is even), selects ${}_2F_1(\alpha, \beta; \kappa; z)$. At $z = 1$ the exponents are 0 and

$\kappa-\alpha-\beta = n/2-1-s$. Since s is imaginary and non-zero, $\kappa-\alpha-\beta$ is not an integer, and Gauss's connection formula [20, Eq. 15.8.4]

$$ {}_2F_1(\alpha,\beta;\kappa;z) = \frac{\Gamma(\kappa)\Gamma(\kappa-\alpha-\beta)}{\Gamma(\kappa-\alpha)\Gamma(\kappa-\beta)}\,{}_2F_1(\alpha,\beta;\alpha+\beta-\kappa+1;1-z) + (1-z)^{\kappa-\alpha-\beta}\,\frac{\Gamma(\kappa)\Gamma(\alpha+\beta-\kappa)}{\Gamma(\alpha)\Gamma(\beta)}\,{}_2F_1(\kappa-\alpha,\kappa-\beta;\kappa-\alpha-\beta+1;1-z) $$

applies. The second coefficient is $C(n,s)$. The Gamma function has no zeros, and α, β and $\alpha+\beta-\kappa$ are not poles, because their imaginary parts are $\mu\omega/2$, $\mu\omega/2$ and $\mu\omega$, all non-zero. □

At 40 digits the formula satisfies the original first-order equations to 10^{-40} for $n = 18$–1000. For the fundamental at $\omega=\omega_\infty$, $|C| = 0.0121$, 0.0114, 0.0124, 0.0192, 0.0405 at $n = 18$, 38, 58, 200, 1000: it grows like $\sqrt{n}$ and is small only through a factor of order $e^{-\pi\mu\omega/2}$. Without gravity a free wave cannot be both regular at the centre and analytic at the light cone. The branch $(1-x^2)^{n/2-1-s}$ is smaller than any power of $1/n$ at every fixed $x<1$, which is why Proposition 1 never sees it; gravity, through the $O(1/n)$ couplings, has to cancel its coefficient harmonic by harmonic. The next two subsections show that it cannot do so inside $0<x<1$, and locate where the cancellation must happen instead.

3.3. **What gravity does to the non-analytic piece.** The non-analytic piece is a fast WKB mode: in the exact solution of Proposition 2 it is $x^{2-n}(1-x^2)^{n/2-1-s}$, which varies on the scale $1/n$. Put

$$ \Phi(x) = \ln x - \tfrac{1}{2}\ln(1-x^2). \tag{3.5} $$

Proposition 3. *Let $F(\theta)$ and ω be given, and $c = 1-\varepsilon F^2(1-x^2)+O(\varepsilon^2)$ as in Proposition 1. The perturbations*

$$ \delta p = e^{-n\Phi(x)}\big(A(x,\theta)+O(1/n)\big), \qquad \delta m = e^{-n\Phi(x)}\Big(\frac{1-x}{1+x}\,A(x,\theta)+O(1/n)\Big) $$

solve the equations linearized about the formal solution on $0<x<1$, to the order shown, if and only if

$$ x(1-x^2)\,\partial_x A - 2\omega x^2\,\partial_\theta A = \big(1+x-2x^2F(\theta)^2\big)\,A, \tag{3.6} $$

that is, along the characteristics $\theta=\theta_0+\omega\ln(1-x^2)$,

$$ A = A_0(\theta_0)\,\frac{x}{1-x}\,\exp\Big(-\int_0^x \frac{2t\,F\big(\theta_0+\omega\ln(1-t^2)\big)^2}{1-t^2}\,\mathrm{d}t\Big). \tag{3.7} $$

Proof. Substitute into the linearized equations. The terms of order n come from $x(x\mp c)\,\partial_x$ acting on $e^{-n\Phi}$ and from $\frac{n}{2}c\,(\delta p+\delta m)$. With B the amplitude of δm they read

$$ -x(x-1)\,\Phi'\,A = \tfrac{1}{2}(A+B), \qquad -x(x+1)\,\Phi'\,B = -\tfrac{1}{2}(A+B). $$

A non-zero solution needs $x(1-x^2)\,\Phi'=1$, which gives Φ, and then $B = A(1-x)/(1+x)$. At order 1 the corrections (a_1,b_1) satisfy a linear system with the same singular matrix,

$$ \frac{1}{2}\begin{pmatrix} \frac{1-x}{1+x} & -1 \\ 1 & -\frac{1+x}{1-x}\end{pmatrix}\begin{pmatrix} a_1 \\ b_1\end{pmatrix} = \begin{pmatrix} R_p \\ R_m\end{pmatrix}, $$

whose left null vector is $(1+x,-(1-x))$. The right-hand sides collect $x(x\mp1)\,\partial_x$ of A and B, the $\omega x\,\partial_\theta$ terms, the $\mp c$ terms, and the gravity terms from $c=1-\varepsilon G_1$ inside $x(x\mp c)\,n\Phi'$ and $\frac{n}{2}c$. The perturbations of b and c, and $G\,\delta p$, are smaller by $1/n$ and drop out. Solvability, $(1+x)R_p-(1-x)R_m=0$, multiplied by $(1+x)/2$, is (3.6); with $G_1 = F^2(1-x^2)$ the gravity terms add up to $-2x^2F^2A$. A first-order equation integrates along its characteristics, and $\int(1+x)/(x(1-x^2))\,\mathrm{d}x = \ln\big(x/(1-x)\big)$ gives the prefactor. □

Without gravity a harmonic $A=\alpha(x)e^{\mathrm{i}\mu\theta}$ gives $\alpha = x(1-x)^{-1}(1-x^2)^{-s}$ and $\delta p-\delta m = 2x^{2-n}(1-x^2)^{n/2-1-s}$: exactly the second solution of Proposition 2, with its exponents at the centre and at the light cone. A symbolic repetition of the derivation from the exact equations, with δb and δc kept, confirms that they enter only at order $\varepsilon^{3/2}e^{-n\Phi}$, and direct integration of the linearized equations reproduces the transported A with relative errors falling like $1/n$ (0.57 at $n=100$ to 0.031 at $n=1600$, for $F = 2.72\sin\theta + 0.22\sin3\theta$, $\omega=\omega_\infty$). Three consequences:

(i) Gravity acts on the non-analytic piece at order 1, not $1/n$ (for the profile above it multiplies it by about 40 over $0.72 \le x \le 0.9$, the interval of the direct integration). Near the light cone the characteristic winds through all phases and the factor behaves like $(1-x^2)^{\langle F^2\rangle}$, so the light-cone exponent becomes $n/2-1-s+\langle F^2\rangle$. With $\langle F^2\rangle \approx 3.7$, an expansion in the amplitude of the wave is not a route to the selection.
(ii) The factor in (3.7) never vanishes, so gravity cannot cancel the fast mode on the way to the light cone.
(iii) Every characteristic reaches $x=0$, where the fast mode is the singular branch x^{2-n}. A solution regular at the centre therefore has $A_0=0$ on every characteristic: at any order of this WKB expansion it carries no fast mode on $0<x<1$. The non-analytic piece must come from a Stokes phenomenon.

3.4. **The light-cone coefficient as a Stokes multiplier.** A divergent series records the exponentially small terms it switches on in its late orders [19, 12]: if the switched-on term is $n^{-g}e^{-n\chi(x)}$, the coefficients behave like $y_k \sim K\,Q(x)\,\Gamma(k-g)/\chi(x)^{k-g}$. At leading order this can be made exact, because the $1/n$ expansion of $x\,{}_2F_1$ is explicit.

Proposition 4. *Let $0<x<1$, Φ as in (3.5), $\tau_\ell = \Phi - \mathrm{i}\pi(\ell+\frac12)$ for $\ell\in\mathbb{Z}$, and let $y = x\,{}_2F_1(\alpha,\beta;\kappa;x^2)$ with α,β,κ as in (3.2), $y\sim\sum_k y_k\varepsilon^k$. If $|\Phi|<(\sqrt3/2)\pi$ (equivalently $0.066<x<0.998$), then*

$$y_k = K\,x^2(1-x^2)^{-1-s}\,\frac{\Gamma(k+s+\frac12)}{\tau_0^{\,k+s+\frac12}}\,\big(1+O(1/k)\big), \qquad K = e^{-\mathrm{i}\pi(1+s)}\,\frac{2^{-s-1/2}}{\Gamma(\alpha)\Gamma(\beta)}. \tag{3.8}$$

The $\ell=-1$ singularity contributes at the same rate, with a k-independent relative weight of 8×10^{-5} ($x=0.5$) to 5×10^{-8} ($x=0.95$). If $|\Phi|>(\sqrt3/2)\pi$, the late orders are governed instead by the x-independent singularities $\pm\mathrm{i}\pi$.

The proof uses one elementary lemma.

Lemma 3.3 (product)**.** *Let $A=\sum_k a_k\varepsilon^k$ and $B=\sum_j b_j\varepsilon^j$ be Gevrey-1 series whose Borel transforms have radii of convergence $\rho_A<\rho_B$, with a_k of exact factorial growth $|a_k|\asymp k!\,\rho_A^{-k}$ (up to powers of k). Then $(AB)_k = b_0a_k\,(1+O(1/k))$.*

Proof. $|b_ja_{k-j}/a_k| \lesssim (\rho_A/\rho_B)^j\,j!\,(k-j)!/k!$, and the sum over $j\ge1$ of the right-hand side is $O(1/k)$. □

Proof of Proposition 4. (i) Euler integral. For $\mathrm{Re}\,\kappa > \mathrm{Re}\,\beta$ (that is, $n>1$), ${}_2F_1 = \frac{\Gamma(\kappa)}{\Gamma(\beta)\Gamma(\kappa-\beta)}\int_0^1 t^{\beta-1}(1-t)^{\kappa-\beta-1}(1-zt)^{-\alpha}\,\mathrm{d}t$. With $1-t=e^{-2\tau}$ and $\kappa=(1+1/\varepsilon)/2$,

$$y = x\,P(\kappa)\cdot 2\int_0^\infty e^{-\tau/\varepsilon}G(\tau)\,\mathrm{d}\tau, \qquad G = e^{(2\beta-1)\tau}(1-e^{-2\tau})^{\beta-1}(1-z+ze^{-2\tau})^{-\alpha},$$

with $P=\Gamma(\kappa)/(\Gamma(\beta)\Gamma(\kappa-\beta))$ and $z=x^2$.

(ii) Singularities of G and Watson's lemma. Write $G=\tau^{\beta-1}H$. H is analytic except where $1-z+ze^{-2\tau}=0$, i.e. at $\tau=\tau_\ell$, each a simple zero, so that $H\propto(1-\tau/\tau_\ell)^{-\alpha}$ there; and at the branch points $\tau=\mathrm{i}\pi m$, $m\neq0$, of $(1-e^{-2\tau})^{\beta-1}$. On $\tau>0$, $1-z+ze^{-2\tau}>1-z>0$ and $|G|\le Ce^{\tau}$, so Watson's lemma gives $2\int_0^\infty e^{-\tau/\varepsilon}G\sim 2\sum_m H_m\Gamma(m+\beta)\varepsilon^{m+\beta}$ with $H=\sum_m H_m\tau^m$.

(iii) Darboux. The radius of convergence of $\sum_m H_m\tau^m$ is $R=\min(|\tau_0|,\pi)$. If $|\tau_0|<\pi$, the only singularities on $|\tau|=R$ are τ_0 and τ_{-1}, both of type $(1-\tau/\tau_\ell)^{-\alpha}$, so

$$H_m = \sum_{\ell=0,-1} h_\ell\,\frac{\Gamma(m+\alpha)}{\Gamma(\alpha)\,m!}\,\tau_\ell^{-m}\,\big(1+O(1/m)\big), \qquad h_\ell = \lim_{\tau\to\tau_\ell} H(\tau)\,(1-\tau/\tau_\ell)^{\alpha},$$

with H continued along the ray from 0.

(iv) Prefactor. $P(\kappa)(2\varepsilon)^\beta(1+\varepsilon)^{-\beta}\Gamma(\beta) = 1+\sum_j p_j\varepsilon^j$, from Binet's expansion of $\ln\Gamma(\kappa)-\ln\Gamma(\kappa-\beta)$, whose Borel transform is singular only at $\tau=\mathrm{i}\pi m$; and $(1+\varepsilon)^\beta$ converges.

(v) Product. By Lemma 3.3, with $b_0=1$ and the powers $\varepsilon^{\pm\beta}$ cancelling, $y_k = \frac{x\,2^{1-\beta}}{\Gamma(\beta)}\,\Gamma(k+\beta)H_k\,(1+O(1/k))$.

(vi) Combining. $\Gamma(k+\alpha)\Gamma(k+\beta)/\Gamma(k+1) = \Gamma(k+s+\frac12)(1+O(1/k))$. Evaluating h_0 in closed form, with $e^{2\tau_0}=z/(z-1)$, the factors reduce to $x^2(1-x^2)^{-1-s}$ times K, which gives (3.8). If $|\tau_0|>\pi$, then $R=\pi$ and step (iii) applies to $\pm\mathrm{i}\pi$ instead. □

The prediction is confirmed numerically to order 122 at 90 digits (`solver/largeD_prop4/`): at $x = 0.9$, y_k divided by (3.8), minus one, extrapolates to $(-0.9-2.3\mathrm{i})\times10^{-3}$ over $k = 100$–122; the measured constant equals K to $1+3.65/k$ at $x = 0.5$, 0.8 and 0.9; and at $x = 0.999$ and 0.9999 the measured singulant is $-3.135\mathrm{i} \to -\mathrm{i}\pi$, not $\Phi - \mathrm{i}\pi/2$, as the statement predicts. The singularities $\pm\mathrm{i}\pi$ come from the Gamma-function prefactor and the endpoint factor $(1-e^{-2\tau})^{\beta-1}$; their exponentials $e^{\mp\mathrm{i}\pi n}$ are pure phases, not exponentially small terms, so they do not enter the Stokes-line argument, which concerns the τ_ℓ.

Remark 3.4 (the Stokes line). Put $\chi = \tau_0 = \Phi - \mathrm{i}\pi/2$. Then $e^{-n\chi} = e^{\mathrm{i}n\pi/2}x^{-n}(1-x^2)^{n/2}$ is the fast mode of Proposition 3, and $x^2(1-x^2)^{-1-s}$ is its transported amplitude at $F = 0$: the divergence of the $1/n$ series and the non-analytic piece are the same object. On the branch continued from $0 < x < 1$, χ vanishes only at $x = \infty$ ($\Phi(\infty) = \mathrm{i}\pi/2$), so the fast mode is born at the outer singular point of the equations. Its Stokes line ($\operatorname{Im}\chi = 0$, $\operatorname{Re}\chi > 0$) is the real axis $x > 1$ approached from above, where $\chi = \ln\left(x/\sqrt{x^2-1}\right)$. On $0 < x < 1$, $\operatorname{Im}\chi = -\pi/2$, so nothing is switched on there, as consequence (iii) of Proposition 3 requires. (On the imaginary axis $\operatorname{Im}\chi = 0$ too, but there $\operatorname{Re}\chi < 0$.) The Stokes line ends at the light cone, so a loop around $x = 1$ crosses it, and that is where the branch $(1-x^2)^{n/2-1-s}$ and its coefficient come from: $C(n,s)$ is the Stokes multiplier. Directly, the $1/n$ series truncated near its smallest term reproduces $x\,{}_2F_1$ inside the light cone far below the size of the singular branch: at $x = 0.9$, $n = 60$ the smallest difference is 4×10^{-40}, against $|C|$ times the branch, 7×10^{-21} (`solver/largeD_linear_truncation.py`).

4. The n-free inner problem and the selection rule

4.1. **The inner equations.** Near $x = \infty$ the large-n expansion is not uniform. At $x \sim \sqrt{n}$ one has $\varepsilon x^2F^2 = O(1)$, so $c - 1 = O(1)$ and gravity is no longer weak; $p \approx \varepsilon^{1/2}xF = O(1)$; and $e^{-n\Phi} \to e^{\mp\mathrm{i}n\pi/2}e^{-n/(2x^2)}$, so fast and slow modes are no longer separated. Put

$$\xi = x/\sqrt{n}, \qquad y = p - m, \qquad \sigma = \sqrt{n}\,(p+m), \qquad b - 1 = \eta/n, \qquad L = \xi^2\partial_\xi + \omega\xi\,\partial_\theta. \tag{4.1}$$

At leading order the difference and the sum of (2.2)–(2.3) and the equations (2.1) become

$$L\,y = c\,\sigma, \qquad L\,\sigma = c\,\xi\,\partial_\xi y + c\left(\tfrac12 y^2 - 1\right)y, \qquad \xi\,\partial_\xi c = \tfrac12\,c\,y^2, \qquad \eta = \tfrac12 y^2. \tag{4.2}$$

(In the difference equation the terms of order $\sqrt{n}$ are $x^2\partial_x y$, $\omega x\,\partial_\theta y$ and $nc\,(p+m)$; in the sum the terms of order 1 are $x^2\partial_x(p+m)$, $xc\,\partial_x y$, $\omega x\,\partial_\theta(p+m)$ and $(G-c)\,y$ with $G = c\eta$; the b equation becomes algebraic, $\eta = p^2 + m^2$. Lemma B5.0 below gives the corrections exactly.) No large parameter is left: the inner problem is the same for every n. Up to a change of variables it is the leading-order flat-space system of Rozali and Way [44, Sec. 2.2], in the regime $r = O(1)$, $t = O(D^{-1/2})$ where they rescale time by $\sqrt{D}$; the map is written out in the proof of Proposition 5(i) and verified symbolically (`solver/inner_pert/rw_map.py`).

The formal solution regular at $\xi = 0$,

$$y = \sum_{j\ge0} d_j\,\xi^{2j+1}, \qquad \sigma = \sum_{j\ge0} s_j\,\xi^{2j+2}, \qquad c = \sum_{j\ge0} g_j\,\xi^{2j}, \qquad d_0 = 2F, \quad g_0 = 1, \tag{4.3}$$

is unique once F is given (the coefficients follow by recursion), and it is Gevrey-1 in ξ^2. Its first terms are $g_1 = F^2$, $s_0 = 2(F + \omega F')$, $d_1 = \omega^2F'' + 3\omega F' + 2F - 2F^3$: the top-degree coefficients of G_1, S_1 and D_1 in (3.1). In general d_k, $-g_k$ and s_{k-1} are the top-degree coefficients of the outer D_k, G_k and S_k (numerically to 2×10^{-13} at orders 1–10, for $F = 2.72\sin\theta$ and $\omega = \omega_\infty$; `solver/largeD_inner.py`): the inner problem resums the outer series' top-degree terms. The free functions K_k of Proposition 1 multiply x, the lowest degree, so they never reach the inner problem, which contains F and ω only.

4.2. **The linear inner problem.** With $t = 1/\xi$ and the characteristic coordinate $\phi = \theta + \omega\ln t$, one has $L = -\partial_t$ at fixed ϕ, and eliminating σ from (4.2) gives

$$y'' + \left(c^2t - \frac{c'}{c}\right)y' + c^2\Big[\omega\,\partial_\phi y + \left(1 - \tfrac12 y^2\right)y\Big] = 0, \qquad t\,c' + \omega\,\partial_\phi c = -\tfrac12\,c\,y^2, \tag{4.4}$$

with $' = \partial_t$ at fixed ϕ: along each characteristic, a softening anharmonic oscillator with friction c^2t. Without gravity ($c = 1$) and for one harmonic, $w = t^{-s}y_\mu$ obeys Hermite's equation $w'' + tw' + (1+s)\,w = 0$, and $w = e^{-t^2/4}v$ gives Weber's equation $v'' + (\frac12 + s - \frac14 t^2)\,v = 0$. The solution regular at $\xi = 0$ is

$$y_\mu = 2F_\mu\,\xi\,{}_2F_0(\alpha,\beta;;2\xi^2), \qquad \text{i.e.} \qquad d_j = 2^j\,\frac{(\alpha)_j(\beta)_j}{j!}\,d_0, \tag{4.5}$$

and the second solution carries $e^{-1/(2\xi^2)}$, which is the outer fast mode $e^{-n/(2x^2)}$. The Stokes line of ${}_2F_0(\cdot\,;2\xi^2)$ is the real ξ axis, i.e. the real axis $x>1$. The lateral solutions are $y_\mu^\pm = 2F_\mu\,\xi\,X^\alpha U(\alpha,\frac12,X)$, $X=-1/(2\xi^2)$, with y^+ on the upper side of the cut of Kummer's U ($\arg X\to\pi^-$) and y^- on the lower side [20, §§2.11, 13.7]. Their difference on real ξ is $2\pi\mathrm{i}\,K'(s)\,2F_\mu$ times the recessive (fast) solution, with

$$K'(s)=\frac{2^{-s-1/2}}{\Gamma(\alpha)\Gamma(\beta)}=\frac{1}{\sqrt{2\pi}\,\Gamma(1+s)} \tag{4.6}$$

by the duplication formula; this normalization is checked in ball arithmetic to $1\pm3\times10^{-30}$ at two radii and two frequencies (`solver/proof/lin_check.py`). Matching the inner fast mode to the outer one gives the large-n form of the light-cone coefficient of Proposition 2, with $\Lambda=n/2-1-s$:

$$C(n,s)=-2\pi\mathrm{i}\,K'(s)\,(n+1)^{s+1/2}\,e^{-\mathrm{i}\pi\Lambda}\big(1+O(s^2/n)\big)=-\mathrm{i}\,\frac{\sqrt{2\pi}}{\Gamma(1+s)}\,(n+1)^{s+1/2}\,e^{-\mathrm{i}\pi\Lambda}\big(1+O(s^2/n)\big), \tag{4.7}$$

Weber's classical Stokes constant $\sqrt{2\pi}/\Gamma(-\nu_W)$ with $\nu_W=-1-s$ [20, §12.9]. The ratio of the two sides is 1.00014 at $n=10^5$ for $\mu=1$ (1.0013 for $\mu=3$), with $n\,(\text{ratio}-1)\to13.9-7.6\mathrm{i}$, of the size of $|s^2|=14.6$ (`solver/largeD_stokes_C.py`); the asymptotics need $\mu^2\omega^2\ll n$. So, for large n, the light-cone coefficient is the n-independent Stokes constant $K'(s)$ of a ${}_2F_0$, times $(n+1)^{s+1/2}$ and a phase. In the linear case the outer late-order constant K of Proposition 4 and this inner Stokes constant are the same object; this is the exact linear version of the inner–outer matching.

4.3. **The fast mode with gravity, and the multipliers.** With $\delta y=e^{-1/(2\xi^2)}a$ in the linearization of (4.2), the terms of order ξ^{-2} cancel and

$$\xi\,\partial_\xi a+2\omega\,\partial_\theta a=2F^2a, \tag{4.8}$$

which is the large-x limit of (3.6). Its solutions are

$$a(\xi,\theta)=\xi^{2\langle F^2\rangle}\,e^{\Pi(\theta)}\,g(\theta_0),\qquad \theta_0=\theta-2\omega\ln\xi,\qquad \Pi(\theta)=\frac{1}{\omega}\int^\theta\big(F^2-\langle F^2\rangle\big), \tag{4.9}$$

with g an arbitrary 2π-periodic function of the characteristic label θ_0. Accordingly the jump of the inner solution across the Stokes line (real ξ), the difference of the lateral sums along $\arg\xi^2=0^\pm$, has the form

$$[y](\xi,\theta)\sim S(\theta_0)\,e^{-1/(2\xi^2)}\,\xi^{2\langle F^2\rangle}\,e^{\Pi(\theta)}\big(1+O(\xi^2)\big),\qquad \xi\to0^+,\qquad S(\theta_0)=\sum_{\nu\ \text{odd}}S_\nu\,e^{\mathrm{i}\nu\theta_0}. \tag{4.10}$$

In the Borel plane conjugate to $1/\xi^2$ the singularity stays at $\zeta=\frac12$ (without gravity the inner series grows at ratio $d_{j+1}/d_j\sim2j$), and the Dingle exponents become $g_\nu=-(\frac12+\mathrm{i}\nu\omega)+\langle F^2\rangle$: the late orders of label harmonic ν grow like $\Gamma(j-g_\nu)$, each spread over the harmonics $\nu+2k$ by e^Π. Darboux fits with these exponents leave residuals of 5×10^{-8} in the one-harmonic model.

Definition 4.1 (fast-type Stokes multipliers)**.** *For a model with finitely many harmonics, the formal series (4.3) is 1-summable in every direction except* $\arg\xi^2=0$ *(Braaksma's theorem [7]; Lemma B4(M) below). The two lateral sums differ on* $\xi>0$ *by an exponentially small solution of the variational equation, and the coefficients of its first term in the basis* $e^{-1/(2\xi^2)}\xi^{2\langle F^2\rangle}e^{\Pi(\theta)}e^{\mathrm{i}\nu\theta_0}$ *are the multipliers* $S_\nu(F,\omega)$*. For the untruncated problem the multipliers are defined order by order in an amplitude expansion (Section 8.1); a definition directly in a space of* θ*-analytic functions is part of gaps (a) and (b) below.*

The selection conditions are $S_\nu=0$ for every odd ν: one complex condition per odd harmonic, for the unknowns F's odd harmonics (less one phase) and ω, so the count balances. Two features make the problem harder than its scalar precedents. First, gravity couples the fast modes of different harmonics, so the multiplier is a vector indexed by harmonics with a matrix structure. Second, the linear multipliers grow with the harmonic: S_ν^{lin} is proportional to $F_\nu/(\Gamma(\alpha_\nu)\Gamma(\beta_\nu))$, and $1/|\Gamma(\alpha_\nu)\Gamma(\beta_\nu)|\approx47$ for $\nu=1$ and 4.4×10^6 for $\nu=3$, growing like $e^{\pi|\nu|\omega/2}$ with $\pi\omega_\infty/2=6.004$. So $S(\theta_0)$ formally involves F continued to $\operatorname{Im}\theta=\mp\pi\omega/2\approx\mp6$, while F's harmonics decay only like $e^{-1.4|\nu|}$, and a harmonic generated only by the nonlinearity already carries a jump 10^2–10^4 times that of harmonic 1. Truncating harmonics is therefore not a controlled approximation; Section 6.6 explains why the weights nevertheless cancel in the tuned problem.

4.4. **The selection rule.**

Conjecture 1 (selection at $D=\infty$)**.** *For $n=D-2$ let $F_n(\theta)=\sqrt{n}\,\partial_x p(0,\theta)$ and $\omega_n=2\pi/\Delta(D)$ be the profile and the frequency of the critical solution. Assume*

(H1) $F_n\to F_\infty$ uniformly on a strip $|\operatorname{Im}\theta|<\sigma_F$ for some $\sigma_F>0$, and $\omega_n\to\omega_\infty>0$ (the data give $\sigma_F\approx 1.4$, the harmonic decay rate); and

(H2) the formal family of Proposition 1 with data (F_n,ω_n,K_k) is the asymptotic expansion of the solution uniformly on $\delta\le x\le 1-\delta$ (checked: residuals $\propto n^{-3}$ through third order).

Then $S_\nu(F_\infty,\omega_\infty)=0$ for every odd ν, where S_ν are the fast-type Stokes multipliers of the inner problem (4.2). Conversely, a zero of S at which the Jacobian with respect to (F,ω), modulo the phase, is invertible should continue to critical solutions for all large n.

The argument. The inner problem resums the outer series' top-degree terms; the outer series' light-cone exponential is fixed by the inner singularity at $x=\infty$ (exactly so in the linear case, where the outer late-order constant of Proposition 4 is the inner Weber constant); the switched exponential reaches the light cone through an invertible map; and analyticity at the light cone forces $C=0$, hence $S=0$. The map $S\mapsto C$ is, at leading order in $1/n$, the composition of three maps:

(1) *Inner–outer matching* on $1\ll x\ll\sqrt{n}$, where the inner fast mode $e^{-1/(2\xi^2)}\xi^{2\langle F^2\rangle}e^{\Pi(\theta)}S(\theta_0)$ matches the outer WKB mode $e^{-n\Phi}A$ on the branch above the real axis $x>1$ (there $\Phi=\ln(x/\sqrt{x^2-1})+\mathrm{i}\pi/2$).

(2) *The outer transport* of Proposition 3 along $x>1$ with real θ. It has real coefficients while $\varepsilon x^2F^2\ll 1$ and acts along real characteristics as multiplication by a non-vanishing function composed with a shift, so it is invertible on Fourier sequences of at most exponential growth.

(3) *The half-monodromy* at the regular singular point $x=1$. It continues the singular Floquet branch $(1-x)^\Lambda\varphi(\theta-\omega'\ln(1-x))$, $\omega'=\omega/(1-\langle G(1)\rangle)$, from $x>1$ to $x<1$ and multiplies harmonic μ by $e^{-\mathrm{i}\pi\Lambda}e^{\pm\pi\mu\omega'}$: a diagonal map with non-zero entries.

So $C\equiv 0\Leftrightarrow S\equiv 0$ at leading order. In the linear case this is exact (Propositions 2 and 4 with (4.7)).

Where rigour fails.

(a) *Borel summability.* Borel summability of the nonlinear series is not proved. For a fixed number of harmonics it is not needed (Section 6). Recomputed in ball arithmetic, the outer late orders of the one- and three-harmonic truncations show no Borel singularity near the positive real axis; with three harmonics there is an x-independent one near $\zeta\approx -1.2$, which does not obstruct summation along $\mathbb{R}^+$ but may approach the origin as the number of harmonics grows (`proof/B3_borel.md`, §5).

(b) *Uniformity in the harmonic index.* The weights $e^{\pi|\mu|\omega/2}$ grow like $e^{6.0|\mu|}$ while F's harmonics decay only like $e^{-1.4|\mu|}$, so an $O(1/n)$ relative error in a high harmonic could be amplified when it mixes into low harmonics through e^{Π}. The leading-order equivalence holds harmonic by harmonic, but the n at which it becomes accurate might grow with the number of harmonics.

(c) *Uniformity at $x=1$.* The outer WKB amplitude is singular there, $A\propto(1-x)^{-1}$, and must be matched to the Frobenius solution on $1/n\ll|1-x|\ll 1$. The exponents agree, $\Lambda=n/2-1-s+\langle F^2\rangle+O(1/n)$ on both sides, but the matching must be written out.

Section 6 shows how far each gap is closed for a model with finitely many harmonics: (c) holds for each fixed number of harmonics up to a routine block-diagonalization, by a Volterra bound that is proved (Lemma B6′ (Volterra bound)); the inner half of (a) is Braaksma's theorem, and the outer half is replaced by fixed-order truncations, its Borel-sum form being hypothesis (H-σ), which the selection theorem does not use; and (b) is narrower than stated here (Section 6.6).

5. An exact unstable mode at infinite dimension

By Remark 3.2 the growth rate $\lambda=1/\gamma$ of a perturbation $\propto e^{\lambda\tau}$ is, like ω, selected beyond all orders in $1/n$, by the same kind of fast-mode multiplier. In the inner problem, however, some growth rates are exact, because they come from symmetries: $\lambda=0$ (a shift of θ), $\lambda=1$ (a shift of the time of the singularity, $\delta Z=(\xi\partial_\xi+\omega\partial_\theta)Z$ for $Z=(y,\sigma,c)$, which is the $D=\infty$ limit of the gauge mode of the solver), and $\lambda=2$, the value the numerical growth rates approach. This section derives the last.

Write the inner problem in physical inner variables: $\rho = r/\sqrt{n}$, with r the areal radius, and the time t. Then $\xi = \rho/(-t)$, $L = \rho\,\partial_t$ at fixed ρ, and $\xi\partial_\xi = \rho\partial_\rho$ at fixed t. Put

$$s = \frac{\rho^2}{2} = \frac{r^2}{2n}, \qquad y = \sqrt{s}\,Y, \qquad \sigma = \sqrt{2}\,s\,W \tag{5.1}$$

(this s is unrelated to the $s = \mathrm{i}\mu\omega$ of (3.2)).

Proposition 5. *(i) In the variables (5.1) the inner equations (4.2) are*

$$Y_t = c\,W, \qquad W_t = c\Big(Y_s + \frac{Y^3}{4}\Big), \qquad c_s = \frac{c\,Y^2}{4}, \tag{5.2}$$

that is, the nonlinear heat equation $Y_s = c^{-1}\partial_t(c^{-1}\partial_t Y) - Y^3/4$ in which s plays the role of time and t that of space. The system contains no explicit s or t, so it is invariant under s-translation, t-translation, the scaling $(t, s, Y, W) \mapsto (at, a^2 s, Y/a, W/a^2)$ and the reparametrizations $t \mapsto T(t)$, $c \mapsto T'c$.

(ii) Let $(y, \sigma, c)(\xi, \theta)$ be a discretely self-similar solution of (4.2) with frequency ω, regular at $\xi = 0$, with $y = 2F\xi + O(\xi^3)$ and $c = 1 + F^2\xi^2 + O(\xi^4)$. Let $P(\theta)$ be the periodic solution of $P + \omega P' = -2F^2$, and $X = \xi\partial_\xi + \omega\partial_\theta$. Then

$$\delta y = e^{2\tau}\Big(\frac{\xi y_\xi - y}{\xi^2} + P\,X y\Big), \quad \delta\sigma = e^{2\tau}\Big(\frac{\xi\sigma_\xi - 2\sigma}{\xi^2} + P\,X\sigma\Big), \quad \delta c = e^{2\tau}\Big(\frac{c_\xi}{\xi} + (P + \omega P')\,c + P\,X c\Big) \tag{5.3}$$

solves the inner equations linearized about (y, σ, c): it is an exact perturbation with growth rate $\lambda = 2$. It is regular at $\xi = 0$ (δy odd and δc even in ξ), it keeps $c = 1$ at the centre ($\delta c = 0$ at $\xi = 0$), and its leading profile is $\delta y = 2f\xi e^{2\tau} + O(\xi^3)$ with

$$f = (2 + \omega\partial_\theta)(F + \omega F') - 2F^3 + P\,(F + \omega F'). \tag{5.4}$$

(iii) Wherever the fast-type Stokes multipliers of the background are defined (by Borel summation; gap (a)), the multipliers of this perturbation are the image of the background's under a linear map, and they vanish when those vanish. So if the $D = \infty$ critical solution is selected by Conjecture 1, $\lambda = 2$ is one of its exact growth rates.

Proof. (i) With $\rho\partial_\rho = 2s\partial_s$ the explicit factors of ρ cancel; this is a direct substitution, done symbolically with zero residual (`solver/inner_pert/heat_form.py`). No coefficient depends on s or t, which gives the translations. Under the scaling every term of each equation picks up the same power of a. Under $t \mapsto T(t)$, Y and W are scalars and ∂_t picks up T', which $c \mapsto T'c$ absorbs.

The same system as Rozali and Way's. Their metric is $-(1-\delta)A\,\mathrm{d}t^2 + \mathrm{d}r^2/A + r^2\mathrm{d}\Omega^2$, with $\tau_{\mathrm{RW}} = \sqrt{D}\,t$, $A = 1 - A_0/D$ and $\delta = \delta_0$ at leading order. Their leading-order equations $\partial_r\delta_0 = -2r(\partial_{\tau_{\mathrm{RW}}}\phi_0)^2$ and $\partial_r\phi_0 = r(1-\delta_0)^{-1/2}\partial_{\tau_{\mathrm{RW}}}[(1-\delta_0)^{-1/2}\partial_{\tau_{\mathrm{RW}}}\phi_0]$ are posed at $r = O(1)$ and times of order $D^{-1/2}$, which is the inner region. With their (r, τ_{RW}) identified with our (ρ, t) and their $1 - \delta_0$ with our c^2, they read $\phi_s = c^{-1}\partial_t(c^{-1}\phi_t)$ and $c_s = \phi_t^2/c$. Then $Y = 2\phi_t/c$ and $W = Y_t/c$ satisfy (5.2) identically (`solver/inner_pert/rw_map.py`); the coefficient $1/4$ of the cubic term is not adjusted but comes out of the map.

(ii) A generator of weight k under the scaling, applied to a self-similar solution, gives a perturbation $\propto e^{k\tau}$. With $\tau = -\ln(-t)$, ∂_t at fixed ρ is $e^\tau(\partial_\tau + \xi\partial_\xi)$, which is $e^\tau X$ on a DSS background, and $\partial_s = \rho^{-1}\partial_\rho = e^{2\tau}\xi^{-2}\,\xi\partial_\xi$ at fixed t. So s-translation, $(\delta Y, \delta W, \delta c) = (Y_s, W_s, c_s)$, has weight 2 and gives the first terms of (5.3); for instance $\sqrt{s}\,\partial_s(y/\sqrt{s}) = \rho^{-2}(\rho y_\rho - y)$. At $\xi = 0$ it changes c by $\lim c_\xi/\xi = 2F^2$ per unit of $e^{2\tau}$, which the normalization $c(t, 0) = 1$ forbids. The infinitesimal reparametrization $t \mapsto t - \eta(t)$ with $\eta = e^\tau P(\theta)$ moves the scalars by $\eta\,\partial_t = e^{2\tau}PX$ and moves c by that plus $\eta' c = e^{2\tau}(P + \omega P')\,c$. At $\xi = 0$, where $Xc = 0$, the two changes of c cancel exactly when $P + \omega P' = -2F^2$. Harmonic by harmonic this reads $(1 + \mathrm{i}\nu\omega)P_\nu = -2(F^2)_\nu$, and $1 + \mathrm{i}\nu\omega \neq 0$, so P exists, is unique and is as smooth as F^2. A symmetry maps solutions to solutions, so its generator solves the linearized equations. For regularity, write $y = \sum_j d_j\xi^{2j+1}$ and $c = \sum_m g_m\xi^{2m}$: the s-translation part is $\sum_{k\geq 0} 2(k+1)\,d_{k+1}\,\xi^{2k+1}$ for y and $\sum_{m\geq 0} 2(m+1)\,g_{m+1}\,\xi^{2m}$ for c, and X preserves parity in ξ. The leading coefficient is $2d_1 + 2P(F + \omega F')$, with $d_1 = \omega^2 F'' + 3\omega F' + 2F - 2F^3$, which is (5.4).

(iii) Write the perturbation as $\delta Z = \mathcal{D}Z$, where $\mathcal{D}$ is a first-order differential operator in (ξ, θ) whose coefficients (ξ^{-2}, ξ^{-1}, $P(\theta)$, $P'(\theta)$ and constants, times $e^{2\tau}$) are analytic away from $\xi = 0$. The formal solution regular at $\xi = 0$ with a given leading profile and $\delta c(0) = 0$ is unique (the same recursion as for the background), so the formal series of δZ is $\mathcal{D}$ applied to the background's. Lateral Borel–Laplace sums commute with differentiation and with multiplication by convergent series, so the jump of δZ across the Stokes line is $\mathcal{D}$ applied to the background's jump, which is zero when the background's multipliers vanish. ∎

The perturbation recursion of the repository (gravity included, $\delta c(0) = 0$ imposed, nothing about symmetries built in), run at $\lambda = 2$ from the profile (5.4), reproduces the predicted coefficients to 10^{-180} relative in every harmonic the Galerkin truncation resolves, for $F = 2.72\sin\theta + 0.15\sin 3\theta$, through $j = 11$ at 45 harmonics (`solver/inner_pert/pert_check3.py 45 11`); with $\lambda = 1.9$ or 2.1 the same f misses by 7.5% at the third order. The modes $\lambda = 0$ and $\lambda = 1$ pass the same kind of check, to 6×10^{-30} and (with gravity) 10^{-177}. The inner fast mode $e^{-1/(2\xi^2)} = e^{-t^2/(4s)}$ is the heat kernel.

The mode is physical, not gauge: s-translation shifts r^2, with r the areal radius, which gives $r_{\mathrm{BH}}^2 \propto p - p^*$, the discretely self-similar counterpart of Rozali and Way's estimate $\gamma = 1/2$ [44]; the physical reading is in [1].

Corollary 1. *Suppose that Conjecture 1 holds, that the $D = \infty$ critical solution has no unstable mode other than $\lambda = 2$ and the gauge mode $\lambda = 1$, and that $\lambda(D)$ is continuous at $1/n = 0$. Then $\gamma_\infty = 1/2$.*

Proof. By Proposition 5(iii) and Conjecture 1, $\lambda = 2$ is a growth rate of the $D = \infty$ critical solution, and by the second hypothesis it is its only physical unstable mode. By continuity the growth rate $\lambda(D)$ of the unstable mode tends to 2, and $\gamma = 1/\lambda \to 1/2$. □

At finite D the outer region and the $1/n$ corrections to the inner equations depend on r explicitly (Lemma B5.0), so the symmetry is broken and the mode survives as an ordinary unstable mode with $\lambda = 2 + O(1/n)$, whose $1/n$ coefficient is measured in [1]. Two questions are open: whether $\lambda = 2$ is the only physical unstable mode at $D = \infty$, and that coefficient, which by first-order perturbation theory is $-S_{1/n}/\partial_\lambda S$ at $\lambda = 2$, with S the perturbation's fast-mode multiplier and $S_{1/n}$ the part induced by the $O(1/n)$ inner equations. The one-harmonic model of Section 7 has a root $\lambda = 1.765$; this is not a counterexample, because its background is not the $D = \infty$ solution and because Galerkin truncation does not commute with the gauge step, which multiplies by $P(\theta)$.

6. The selection theorem programme

This section sets out a proof of the selection rule for models with finitely many harmonics, after Xie and Tanveer [49, 48], with inner–outer matching after Baldomá, Fontich, Guardia and Seara [5]. It states what is proved, what is cited, what is certified for one harmonic, and what remains. The details are in the repository's `proof/C_selection.md`, `proof/B3_borel.md`, `proof/B5_inner.md` and `proof/B8_constants.md`; uniformity in the number of harmonics is discussed in Section 6.6. Throughout this section $z = \xi^2 = x^2/n$.

6.1. The M-harmonic model and the theorem.

Definition 6.1. *For odd M let P_M keep the harmonics $e^{\mathrm{i}\mu\theta}$ with odd $|\mu| \le M$ for p, m and even $|\mu| \le M+1$ for b, c. The M-*harmonic model *is (2.1)–(2.3) with P_M applied to every right-hand side. For real fields it is a finite analytic system of ODEs in x, with $4N + 2(2N+1) = 8N+2$ real unknowns, $N = (M+1)/2$.*

The even harmonics go up to $M+1$, not $M-1$, because that is the truncation of the inner problem solved in Section 7 and by the inner solvers: with $|\mu| \le M-1$ the inner limit of the model would be a different problem (c on $\{0\}$ instead of $\{0, \pm 2\}$ for $M = 1$). The certificates of Lemmas B5 and B8 below project every binary product onto the harmonics of its result; for $M = 1$ this is the same as projecting each right-hand side once, for $M \ge 3$ it is not.

The solvers of [1] are not such truncations: the double-precision solver works on a Fourier grid in τ with a cut-off, and the double-double solver uses collocation in τ with no spectral truncation of the products. Both discretize the full problem, which the M-harmonic models approximate as M grows, so the M-harmonic model is a model of the computed solutions, not the discrete system actually solved. P_M commutes with ∂_θ and preserves the parity classes, which is all that the induction of Proposition 1 uses.

Theorem B(M). *Fix odd M and let (F_*, ω_*) be a zero of the inner multipliers, $S_\mu(F_*, \omega_*) = 0$ for all odd $\mu \le M$, at which the Jacobian of $(\operatorname{Re} S_\mu, \operatorname{Im} S_\mu)$ with respect to (F's harmonics, ω), with the phase of F_1 fixed, is invertible. Assume Lemmas B3(M), B5(M) and B6(M) below in C^1 form, and the global continuation Lemma B8(M). Then:*

(1) Characterization. *For $n \ge n_0$, the centre-regular solution with data (F_n, ω) near (F_*, ω_*) is analytic at the light cone exactly when, for all odd $\mu \le M$,*

$$C_{n,\mu}(F_n, \omega) = \kappa_{n,\mu}\left[S_\mu(F_n^{(0)}, \omega) + R_{n,\mu}(F_n, \omega)\right] = 0, \tag{6.1}$$

TABLE 2. The lemmas of Theorem B(M).

Lemma	Content	Status
B1(M)	formal $1/n$ solution for every (F,ω)	proved
B2(M)	slow–fast structure; turning point at $x=\infty$	proved
B3(M)	formal series Gevrey-1; centre-regular solution with fixed-order asymptotics	formal half proved; existence is Lemma B8 on the real segment
(H-σ)	outer series Borel summable along $\mathbb{R}^+$	proved in the linear case and near $\zeta=0$; conjecture otherwise; not used
B4(M)	inner multipliers well defined, analytic in (F,ω)	cited (Braaksma); analyticity proved given Braaksma
B5.0	exact equations = inner problem + corrections in $1/n$	proved
B5.1	fast-mode variable and matching factor	proved
B5(M)(1)	sectorial identification on $\mathcal{M}^{\pm}$	proved given (H8); certified for $M=1$, $n\geq e^{73.89}$
B5(M)(2)	outer jump $=\kappa[S+o(1)]$	proved given B5(M)(1); for $M=1$ the error constant is only estimated
B6′, B6(M)	light-cone matching; Volterra bound	proved, up to a routine block-diagonalization
B8.1, B8.2	progressive paths; Stokes geometry	proved
B8(M)	global continuation	proved; certified for $M=1$, $n\geq e^{62}$, and (H8) for $n\geq e^{73.89}$

where $\kappa_{n,\mu}\neq 0$ is explicit, $R_{n,\mu}=O(1/n)$ in C^1 uniformly near (F_*,ω_*), and $F_n^{(0)}$ is the leading part of the profile (the corrections K_k enter only through R). In the linear case $\kappa_{n,\mu}\propto -2\pi\mathrm{i}\,(n+1)^{s+1/2}e^{-\mathrm{i}\pi\Lambda_\mu}$, $s=\mathrm{i}\mu\omega$, as in (4.7), times the transport and matching factors.

(2) Existence and uniqueness of the branch. *There is a unique solution (F_n,ω_n) of these equations near (F_*,ω_*), with $F_n-F_*=O(1/n)$ and $\omega_n-\omega_*=O(1/n)$. It gives a discretely self-similar solution of the M-harmonic model, regular at the centre and analytic at the light cone.*

(3) Converse. *If a sequence of such solutions has $(F_n,\omega_n)\to(F_\infty,\omega_\infty)$, then $S(F_\infty,\omega_\infty)=0$.*

Proof given the lemmas. The first equality in (6.1) is the chain: the light-cone coefficient equals $k_{n,\mu}$ times the coefficient of the fast mode just inside $x=1$ (Lemma B6); this equals $k_{n,\mu}$ times the transport factor times the fast-mode content generated on $x>1$ (Proposition 3 and the half-monodromy); that content is the outer Stokes jump on the line $x>1$ (Lemmas B3 and B8); and the outer jump is the matching factor times $[S_\mu+O(1/n)]$ (Lemma B5 with Lemma B4). Every factor is non-zero and explicit at leading order, and the $O(1/n)$ is uniform in C^1. Divide by $\kappa_{n,\mu}$. The map $Z\mapsto S(Z)+R_n(Z)$ on a neighbourhood of $Z_*=(F_*,\omega_*)$ in $\mathbb{R}^{2N}$ is C^1-close to S, which has an invertible derivative at Z_*, so the finite-dimensional implicit function theorem (or a quantitative inverse-function theorem, since $\|R_n\|_{C^1}=O(1/n)$) gives a unique zero Z_n with $|Z_n-Z_*|=O(1/n)$. The converse follows by continuity: $S(Z_n)=-R_n(Z_n)=O(1/n)\to 0$. □

The correction is $O(1/n)$, not $O(n^{-1/2})$, because the inner variable is $\xi^2=x^2/n$ and the outer equations depend on n only through n and x^2 (Lemma B5.0). The non-degeneracy hypothesis is what layer A certifies (Section 7 for $M=1$); at $M=5$ the reduced 2×2 Jacobian has singular values about 6.8×10^3 and 3.9×10^3 numerically.

Table 2 summarizes the status of the lemmas.

6.2. **The proved lemmas.**

Lemma B1(M). *For every smooth F with odd harmonics $|\mu|\leq M$, every ω_0 and every choice of the free data (ω_k and K_k, also truncated), the M-harmonic model has exactly one formal solution with the structure of Proposition 1.*

Proof. Repeat the induction of Proposition 1 with P_M applied to each right-hand side. Steps (1)–(5) use only algebraic operations and ∂_θ, integration in x, and the parity of x-polynomials; P_M acts on the θ-dependence only, commutes with all of these, and maps even (odd) x-polynomials with periodic coefficients to the same class. Step (6) holds because P_M preserves the odd and even harmonic classes. □

Write the system as $Y_x=n\,\mathcal{F}_1(x,Y)+\mathcal{F}_0(x,Y,\partial_\theta Y)$ with $Y=(p,m,b,c)$ and $\mathcal{F}_1$ collecting the terms of order n. The slow manifold of the formal solution is $b=1+O(\varepsilon^2)$, $p+m=O(\varepsilon^{3/2})$.

Lemma B2(M). *On the slow manifold, the Jacobian of $\mathcal{F}_1$ with respect to (p, m, b, c) has, for each harmonic, the eigenvalues*

$$\lambda_u = \frac{c^2}{x\,(x^2 - c^2)}, \qquad \lambda_b = -\frac{1}{x}, \qquad 0, \qquad 0.$$

The λ_u eigenvector is the combination $u = p+m$ (with a b-component), the λ_b eigenvector is $b-1$, and $v = p-m$ and c are slow. At $c = 1$, $\lambda_u = -\Phi'(x)$, the singulant of Proposition 3.

Proof. The terms of order n contain no θ-derivative, so at leading order they act diagonally on harmonics. Read the four right-hand sides off the equations as functions of (p, m, b, c), take the Jacobian, divide by n, let $n \to \infty$ and set $b = 1$, $m = -p$. The result is the list above (computed symbolically, `solver/proof/fast_eigenvalues.py`). □

Consequently there are no turning points on the real axis; $x = 0$ and $x = c(1) = 1$ are doubly singular points of the type $\varepsilon\,(x - x_0)\,y' = f$; and $x = \infty$ is a turning point, because $\lambda_u \sim n/x^3$ and $\lambda_b = -n/x$ fall to the size $\sim 1/x$ of the slow terms at $x \sim \sqrt{n}$. The inner problem is the turning-point equation of this singular perturbation problem, as in Xie and Tanveer's finger and Kruskal and Segur's dendrite. The fast block is λ_u times the identity on the N odd harmonics at leading order; the transport and the gravity coupling lift the degeneracy at order 1.

Lemma B5.0. *In the inner variables $x = \sqrt{n}\,\xi$, $y = p - m$, $\sigma = \sqrt{n}\,(p + m)$, $b = 1 + \eta/n$, with $\nu = n^{-1/2}$ and the $p - m$ equation multiplied by ν, the exact equations (2.1)–(2.3) become*

$$\mathcal{I}(Z) + \nu^2\,\mathcal{Q}_1(Z) + \nu^4\,\mathcal{Q}_2(Z) = 0, \qquad Z = (y, \sigma, \eta, c),$$

where $\mathcal{I}$ is the inner problem and $\mathcal{Q}_1$, $\mathcal{Q}_2$ are polynomials in Z and its first derivatives; for instance the c equation is $\xi c_\xi - c\eta + \nu^2 c\eta = 0$ and the $p + m$ equation is $\xi^2\sigma_\xi + \omega\xi\sigma_\theta - c\,\xi y_\xi + c\,(1 - \eta)\,y + \nu^2 c\eta y = 0$. Only even powers of ν occur, and the same holds for the M-harmonic model.

Proof. Direct substitution, done symbolically (`solver/proof/inner_corrections.py`); projection onto $|\mu| \le M$ commutes with every step. □

With $\eta = \frac{1}{2}y^2$ the leading equations are (4.2). The η equation is singularly perturbed (its ν^2 term contains $\xi\eta_\xi$): this is the b fast mode $\eta \sim x^{-n}$, which is exponentially small in every direction for $|x| > 1$, so η is slaved to y, $\eta = \frac{1}{2}y^2 + n^{-1}\eta_1 + O(n^{-2})$, and takes part in no Stokes phenomenon. A solution of the inner problem therefore solves the exact equations up to $n^{-1}\mathcal{Q}_1 + O(n^{-2})$, and the multipliers receive corrections in integer powers of $1/n$, the first from the linearized inner problem forced by $\mathcal{Q}_1$.

For $|x| > 1$ put $w(x) = -\frac{1}{2}\ln(1 - x^{-2})$.

Lemma B5.1. *(1) Continuing Φ from $0 < x < 1$ to $x > 1$ through the upper or the lower half plane gives $\Phi = w \pm \mathrm{i}\pi/2$, so $e^{-n\Phi} = e^{\mp \mathrm{i}n\pi/2}e^{-nw}$.*

(2) With $x^2 = nz$, $nw = \frac{1}{2z} + \frac{1}{4nz^2} + \frac{1}{6n^2z^3} + \dots$, so $e^{-nw} = e^{-1/(2z)}\big(1 + O(1/(n|z|^2))\big)$.

(3) The fast mode of Proposition 3 contributes to y and σ the factors $-2x^2/(x^2 - 1) = -2(1 + \frac{1}{nz} + \dots)$ and $-2\sqrt{n}\,x/(x^2 - 1) = -2z^{-1/2}(1 + \frac{1}{nz} + \dots)$.

(4) Along the characteristic the label continues as $\theta_0 = \theta - \omega\ln(1 - x^2) = \theta - 2\omega\ln\xi - \omega\ln n \pm \mathrm{i}\pi\omega + O(\frac{1}{nz})$, and the gravity factor contains $(1 - x^2)^{\langle F^2\rangle} = e^{\mp \mathrm{i}\pi\langle F^2\rangle}n^{\langle F^2\rangle}z^{\langle F^2\rangle}(1 - \frac{\langle F^2\rangle}{nz} + \dots)$.

Hence, in inner variables, the outer fast mode with label harmonic μ, in the gauge $\tilde{A}_0 = A_0 e^{-\Pi}$, is the inner fast mode $e^{-1/(2z)}z^{\langle F^2\rangle}e^{\Pi(\theta)}e^{\mathrm{i}\mu\theta_0^{\mathrm{in}}}$ times

$$\kappa^{\pm}_{n,\mu} = -2\,e^{\mp \mathrm{i}n\pi/2}\,e^{\mp \mathrm{i}\pi\langle F^2\rangle}\,n^{\langle F^2\rangle - \mathrm{i}\mu\omega}\,e^{\mp\pi\mu\omega}\left(1 + O\big(\tfrac{1}{n|z|^2}\big)\right), \tag{6.2}$$

which is explicit and non-zero. In the M-harmonic model the scalar $n^{\langle F^2\rangle - \mathrm{i}\mu\omega}$ becomes the invertible matrix $n^{\mathcal{G}_M}$, with $\mathcal{G}_M$ the truncated transport generator. At $F = 0$ the factor is n^{-s} times the σ-normalization $n^{1/2}$ and the half-monodromy phase: the large-n form $(n + 1)^{s+1/2}$ of the linear $C(n, s)$.

Proof. (1) For $x > 1$, $1 - x^2 = (x^2 - 1)e^{\mp \mathrm{i}\pi}$ on the two continuations, so $-\frac{1}{2}\ln(1 - x^2) = -\frac{1}{2}\ln(x^2 - 1) \pm \mathrm{i}\pi/2$ and $\Phi = -\frac{1}{2}\ln(1 - x^{-2}) \pm \mathrm{i}\pi/2$. (2) $nw = -\frac{n}{2}\ln\big(1 - \frac{1}{nz}\big) = \frac{n}{2}\sum_{k\ge1}(nz)^{-k}/k$. (3) and (4) follow from $\delta m/\delta p = (1 - x)/(1 + x)$, from (3.7), and from the same continuation as in (1); the expansions are printed by `solver/proof/inner_corrections.py`. Collecting the factors gives (6.2). □

The relative correction in (2) is $O(1/(n|z|^2))$, small only for $|z| \gg n^{-1/2}$, i.e. $|x| \gg n^{1/4}$; this sets the outer edge of the matching domain of Lemma B5 below.

The global geometry. Fix $\delta \in (0, \frac{1}{4})$, an angle $\vartheta_0 \in (0, \pi/4)$ and $k_1 > \frac{1}{2}$, and put $\alpha_B = \pi/4 + \vartheta_0/2$ and $R_n = (k_1 n/\ln n)^{1/2}$. Define

$$\mathcal{D}_n^+ = \big\{x : 0 < |x| \le R_n,\ 0 \le \arg x \le \alpha_B\big\} \setminus \big\{|x-1| < \delta/2\big\}, \qquad \mathcal{D}_n^- = \overline{\mathcal{D}_n^+}. \tag{6.3}$$

$\mathcal{D}_n^+$ contains the segment $(0, 1-\delta/2]$, an arc around the light cone through the upper half plane, the Stokes line $[1+\delta/2, R_n]$, and, in $z = x^2/n$, the directions $\arg z \in (\pi/2, \pi/2 + \vartheta_0]$ near $|z| = k_1/\ln n$, where $e^{-1/(2z)}$ is exponentially large. It avoids $x = \pm 1$ and the point $x = c(x) \approx n/F^2$ beyond R_n.

Lemma B8.1 (progressive paths). *For $x \in \mathcal{D}_n^+$ let P_x be the path from 0 along the ray $\arg x = \alpha_B$ to radius $|x|$, followed by the arc of radius $|x|$ clockwise to $\arg x$. Then $P_x \subset \mathcal{D}_n^+$, and $\operatorname{Re}\Phi$ is non-decreasing along P_x; so is the real part of the b-mode phase $\ln x$.*

Proof. Write $r = |x|$, $\varphi = \arg x$ and $Q = |1 - x^2|^2 = 1 - 2r^2\cos 2\varphi + r^4$, so that $\operatorname{Re}\Phi = \ln r - \frac{1}{4}\ln Q$. On the ray $\varphi = \alpha_B$, $\partial_r \operatorname{Re}\Phi = (1 - r^2\cos 2\alpha_B)/(rQ) > 0$, since $\cos 2\alpha_B < 0$ (this is where $\alpha_B > \pi/4$ is used). On the arc, $\partial_\varphi \operatorname{Re}\Phi = -r^2 \sin 2\varphi/Q < 0$ for $0 < \varphi < \pi/2$, so $\operatorname{Re}\Phi$ increases as φ decreases from α_B. Since $|re^{\mathrm{i}\varphi} - 1|$ increases with $\varphi \in [0, \pi]$, the arc from α_B down to $\arg x$ stays outside the disc $|x-1| < \delta/2$ whenever its endpoint x does. Finally $\operatorname{Re}\ln x = \ln r$ is constant on the arc and increasing on the ray. □

So every fast Volterra kernel $e^{-n(\Phi(x)-\Phi(x'))}$, with x' before x on P_x, has modulus at most 1; the transport factor contributes at most $|x/x'|^{2\operatorname{Re}\langle F^2\rangle}$.

Lemma B8.2 (Stokes geometry). *On $\mathcal{D}_n^+$ the anti-Stokes curve $\operatorname{Re}\Phi = 0$ runs from $x = 1/\sqrt{2}$ to $\infty e^{\mathrm{i}\pi/4}$, and the only Stokes line is $x > 1$, where $\chi = \Phi \mp \mathrm{i}\pi/2 = w$ is real and positive. Hence the continuation of the centre-regular solution through the upper half plane carries no free fast mode on $\mathcal{D}_n^+$: it has none near the centre, where $e^{-n\Phi}$ is exponentially large, and no Stokes line is crossed before $x > 1$ is reached from above. Its light-cone singularity is the monodromy of that continuation around $x = 1$, and the difference $Y^+ - Y^-$ of the two continuations on $x > 1$ is the Stokes jump; by the half-monodromy it equals $C_n(e^{-\mathrm{i}\pi\Lambda} - e^{\mathrm{i}\pi\Lambda})\phi$, with ϕ the Frobenius branch of Lemma B6.*

The light cone is handled by a Volterra bound. Near $x = 1$, with $t = 1 - x$, the transport of Proposition 3 becomes $tA_t = \mathcal{T}A$ with $\mathcal{T} = -\omega\partial_\theta - 1 + F(\theta)^2$. Here F^2 couples the harmonics at order 1, but the gauge e^{Π} removes the coupling exactly: $(-\omega\partial_\theta + F^2)\psi = \lambda\psi$ has the periodic solutions $\psi = e^{\Pi(\theta)}e^{\mathrm{i}\mu\theta}$ with $\lambda = \langle F^2\rangle - \mathrm{i}\mu\omega$. So, untruncated, the light-cone exponents are

$$\Lambda_\mu = \tfrac{n}{2} - 1 - \mathrm{i}\mu\omega + \langle F^2\rangle + O(1/n), \tag{6.4}$$

and they differ pairwise by the non-zero imaginary amounts $\mathrm{i}(\nu - \mu)\omega$: no two differ by an integer, for any n. In the M-harmonic model $\mathcal{T}$ is replaced by its truncation $\mathcal{T}_0 = P_M\mathcal{T}P_M$, which breaks the gauge step; at the $M = 5$ zero of Section 9.2 and for M up to 25, the central eigenvalues converge to $\langle F^2\rangle - 1 - \mathrm{i}\mu\omega$ exponentially in M, and all pairwise differences have $|\operatorname{Re}| < 0.02$ and $|\operatorname{Im}| \ge 6.89$ ($|\operatorname{Im}| \ge 7.26$ for $M \ge 3$) (`solver/proof/lightcone_exponents.py`).

Lemma B6′ (Volterra bound). *Let $t_0 = \frac{1}{2}$. Split the linearized M-harmonic system on $t \in (0, t_0]$ into fast components a (the $p+m$ direction of Lemma B2, one per odd harmonic) and slow components v ($p-m$, $b-1$ and c, after eliminating $b-1$ along its own fast direction), so that*

$$t\,a' = \big(n\,\ell(t)\,I + \mathcal{T}_0 + t\,\mathcal{T}_1(t)\big)\,a + t\,\mathcal{Q}(t)\,v, \qquad v' = \mathcal{R}(t)\,a + \mathcal{S}(t)\,v,$$

with $\ell(t) = t\,\Phi'(1-t) = 1/((1-t)(2-t))$, $\mathcal{T}_0 + t\mathcal{T}_1$ the order-1 transport, and $\mathcal{Q}, \mathcal{R}, \mathcal{S}$ analytic and bounded uniformly in n. Let U solve $t\,U' = (\mathcal{T}_0 + t\mathcal{T}_1)\,U$ exactly and put $\psi(t) = \int^t \ell(t')\,\mathrm{d}t'/t'$. Then for every $\mathbf{k} \in \mathbb{C}^N$ there is a solution

$$a = e^{n\psi(t)}\,U(t)\,\big(\mathbf{k} + r(t)\big), \qquad v = e^{n\psi(t)}\,O(t/n), \qquad \sup_{0<t\le t_0} |r(t)| \le C|\mathbf{k}|/n,$$

with C depending only on M, F, ω and t_0.

Proof. (1) $\psi' = \ell/t > 0$ and $e^{n\psi} \sim t^{n/2} \to 0$ as $t \to 0$. (2) Take for v the particular solution $v = \int_0^t(\cdots)$, which is fast-type; then $v' = \mathcal{S}v + \mathcal{R}\,e^{n\psi}U(\mathbf{k} + r)$, and one integration by parts in the factor $e^{n\psi}$ gives $v = e^{n\psi}\big[t\,\mathcal{R}U(\mathbf{k}+r)/(n\ell) + O(t/n^2)\big]$, using $\ell \ne 0$; the boundary term at $t = 0$ vanishes. (3) Substituting the

ansatz for a, the transport cancels exactly and $r' = e^{-n\psi}U^{-1}\mathcal{Q}v = t\,U^{-1}\mathcal{Q}\mathcal{R}U\,(\mathbf{k}+r)/(n\ell) + O(t/n^2)$. With $U(t) = t^{\mathcal{T}_0}(I + O(t))$, the entries of $U^{-1}(\mathcal{Q}\mathcal{R})U$ are $O(t^{-\rho})$, where ρ is the largest difference of the real parts of the eigenvalues of $\mathcal{T}_0$: $\rho = 0$ untruncated and $\rho < 0.02$ in the truncations. So $|r'| \le C_2\, t^{1-\rho}(|\mathbf{k}|+|r|)/n$, which is integrable at 0. (4) With $r(t_0) = 0$, Gronwall's inequality gives $|r(t)| \le |\mathbf{k}|\left(e^{C_2 t_0^{2-\rho}/n} - 1\right) \le C|\mathbf{k}|/n$ uniformly up to $t = 0$. □

As $t \to 0$, $a \sim t^{n\ell(0)} t^{\mathcal{T}_0}(\mathbf{k} + r(0))$, the Frobenius branch with $\Lambda_\mu = n/2+$ (eigenvalues of $\mathcal{T}_0$). This gives:

Lemma B6(M) (light-cone matching). *The exact solution of the linearized M-harmonic system that agrees at t_0 with the WKB fast mode of label harmonic μ equals $k_{n,\mu}\phi_\mu$ plus a solution analytic at $x = 1$, where $\phi_\mu = (1-x)^{\Lambda_\mu}(e_\mu + O(1-x))$ is the unique (by non-resonance) singular Frobenius branch, and $k_{n,\mu} = k_\mu^{(0)}(1+O(1/n))$ with $k_\mu^{(0)} \neq 0$ explicit from the transport. In the linear case the identification is exact (Propositions 2–3).*

What remains in Lemma B6(M) is the routine block-diagonalization of the order-n part on $[t_0, 1-\delta]$ by an analytic transformation bounded uniformly in n, which exists because the fast eigenvalue there is simple and non-vanishing (Lemma B2). Because $x - c$ has a simple zero at the light cone, the WKB phase and the Frobenius exponent are the same object, and the argument is Liouville–Green error control up to a regular singular point [38, Chs. 10–12]; for second-order linear equations the existence and Borel summability of exact WKB solutions is in [37].

6.3. Lemmas B3 and B4.

Lemma B3(M). *(i) For every odd M, every (F, ω) and every $\rho > 0$, the formal solution of Lemma B1 is Gevrey-1 uniformly on $|x| \le \rho$, in the gauge $c(0) = 1$ and in the light-cone gauge: there are $C, t_1 > 0$ with $|Y_k(x)| \le C\,k!\,t_1^{-k}$ there, so its Borel transform in ε is analytic on $|\zeta| < t_1$ for every such x.*

(ii) For $n \ge n_0$ the centre-regular solution with profile F_n exists on $(0, 1-\delta/2]$ and differs from the truncated formal series by the fixed-order remainder of Lemma B8.

Part (i) is proved by a majorant argument closed by the implicit function theorem for a positive polynomial system (`proof/B3_borel.md`, Lemma B3.7). Part (ii) is Lemma B8 restricted to the real segment. The results we read on singularly perturbed systems, with a turning point [8] or doubly singular points [9], assume every component fast, while here $p - m$ and c are slow, so neither part is a citation.

For the centre-regular solution to be the 1-sum of the outer series in ε along $\arg\varepsilon = 0$, uniformly on compacts of $(0, 1]$, the Borel transform must continue, with exponential bounds, to a neighbourhood $S_\sigma = \{\zeta : \operatorname{dist}(\zeta, [0,\infty)) < \sigma\}$ of $[0, \infty)$; by Nevanlinna's theorem in Sokal's form [45] that is exactly the condition. We call it hypothesis (H-σ). Theorem B(M) does not use it: Lemmas B5 and B8 work with actual solutions and fixed-order truncations. The condition is on the distance from $[0, \infty)$, not on the argument: the u-mode singulant $\Phi(x) - \mathrm{i}\pi/2$ has argument down to -0.30π at $x = 0.95$, but it lies on $\operatorname{Im}\zeta = -\pi/2$. (H-$\sigma$) is proved in the linear case with $\sigma = \pi/2$, and near $\zeta = 0$ for every M by part (i), which excludes the would-be singularity at $2\Phi(x)$ near $x = 1/\sqrt{2}$, where 2Φ reaches 0. Elsewhere it is supported by a ball-arithmetic computation of the late orders with Padé approximants [39] (`proof/B3_borel.md`, §5).

Lemma B4(M). *For fixed M, F and ω, the inner problem (4.2), written for the harmonic components of $y = \xi\,\mathsf{Y}(z)$, $\sigma = \xi^2\,\Sigma(z)$, $c = \mathsf{C}(z)$, is a finite system $z^2\,\mathrm{d}\mathsf{Z}/\mathrm{d}z = H(z, \mathsf{Z})$ with H polynomial, $z = 0$ being an irregular singular point of Poincaré rank 1. Its formal power-series solution with $\mathsf{Y}(0) = 2F$, $\mathsf{C}(0) = 1$ is 1-summable in every direction except $\arg z = 0$ [7], its only exponential being $e^{-1/(2z)}$. The multipliers $S_\nu(F, \omega)$ of Definition 4.1 are therefore well defined.*

The first equation is $2z\mathsf{Y}_z + (1 + \mathrm{i}\mu\omega)\mathsf{Y} = (\mathsf{C}\Sigma)_\mu$, regular-singular; the c equation $\mathsf{C}_z = \frac{1}{4}(\mathsf{C}\mathsf{Y}^2)$ is regular; and eliminating Y_z from the second equation leaves $2z^2\Sigma_z = (\mathsf{C}^2\Sigma)_\mu + O(z)$. Braaksma's theorem needs only a meromorphic system and a formal power-series solution. Costin's sharper rank-one theorems [16] do not apply as stated, because the fast eigenvalue is repeated across harmonics and the slow directions have eigenvalue 0; they are not needed, because only the formal power series, not a general transseries, is summed. The repeated eigenvalue makes the first exponential term a vector over harmonics: its N complex coefficients are the S_ν, the count the selection needs. Braaksma's theorem is stated for $x^{\nu+1}y' = F(x, y)$ without parameters, and the abstract of Ramis and Sibuya's proof [42] does not address them either. The analytic dependence of the sums, and of the S_ν, on (F, ω), which Theorem B(M) uses, follows from uniqueness on a ray where $e^{-1/(2z)}$ is exponentially large and a contraction there that is uniform in the parameters; the Picard iterates are analytic in (F, ω) and converge uniformly (`proof/B3_borel.md`, Lemma B4.P). For $M = 1$ this contraction is the one certified on a parameter box in Section 7.

6.4. **The two contractions.** Fix $\frac14<\varpi<\frac12$, with ϑ_0 and k_1 as above, and let

$$\mathcal{M}^+=\big\{z: n^{-1+2\varpi}\le|z|\le k_1/\ln n,\ 0\le\arg z\le\pi/2+\vartheta_0\big\},\qquad \mathcal{M}^-=\overline{\mathcal{M}^+},$$

which is $n^{\varpi}\le|x|\le R_n$, $0\le\arg x\le\alpha_B$, a subset of $\mathcal{D}_n^+$. On $\mathcal{M}^\pm$ the corrections of Lemmas B5.0 and B5.1 have relative size at most $n^{1-4\varpi}\to0$. By Braaksma's theorem the inner formal solution has two lateral sums, $Z_{\rm in}^+$ analytic and asymptotic to the series on $-\pi/2<\arg z<5\pi/2$ and $Z_{\rm in}^-$ on $-5\pi/2<\arg z<\pi/2$; $Z_{\rm in}^+$ is characterized by being asymptotic to the series where $e^{-1/(2z)}$ is exponentially large, $\pi/2<\arg z\le\pi/2+\vartheta_0$. Differences are measured in the variables (P,Φ,C,ϑ) defined by $y=\xi P$, $\Phi=2zP_z$, $C=c$, $\eta=z\Theta$, $\vartheta=\Theta-\frac12[PP]$, with weights $m_\Phi(z)=|z|$ and $m_P=m_C=m_\vartheta=1$. The working domain $\mathcal{W}_n^+\supset\mathcal{M}^+$ is $n^{\varpi}\le|x|\le(k_1n/(\cos\vartheta_0\ln n))^{1/2}$, $0\le\arg x\le\alpha_B$.

Hypothesis (H8) (the input from Lemma B8). $Y_{\rm out}^+$ is analytic, and analytic in the parameters, on a neighbourhood of $\mathcal{W}_n^+$; and on the inner arc $|x|=n^{\varpi}$, $0\le\arg x\le\alpha_B$, the difference $W=Y_{\rm out}^+-Z_{\rm in}^+$ satisfies, for every harmonic, $|W_P|\le\delta_P/n$, $|W_\Phi|\le\delta_\Phi n^{-2+2\varpi}$, $|W_C|\le\delta_C/n$ and $|W_\vartheta|\le\delta_\vartheta/n$. At the upper end of the arc this is the quantitative form of "no exponentially large fast content".

Lemma B5(M). *Let $Y_{\rm out}^\pm$ be the continuations of the centre-regular solution through the upper and lower half planes, written in inner variables.*

(1) Sectorial identification. *Assume (H8). Then on $\mathcal{W}_n^\pm\supset\mathcal{M}^\pm$, $|W_j(z)|\le x_j\,m_j(z)/n$ for every component j, where x is any positive solution of an explicit linear inequality $b+Lx<x$ (componentwise), with b the forcing plus the data δ of (H8). Such an x exists for $n\ge n_0(M)$.*

(2) The jump. *On $\arg z=0$, $Y_{\rm out}^+-Y_{\rm out}^-$ is a fast-type solution of the variational equation whose label function has μ-th coefficient $J_{n,\mu}=\kappa_{n,\mu}\big[S_\mu(F,\omega)+O\big(n^{-1+1/(2k_1)}(\ln n)^{\mathrm{Re}\langle F^2\rangle}\big)\big]$, uniformly in C^1.*

Part (1) is proved by a bootstrap rather than a contraction mapping, because the b-mode needs paths along which $|z|$ does not decrease while the fast mode needs progressive ones (ray $\arg z=\pi/2+\vartheta_0$, then an arc); the data come from (H8) on the whole inner arc, no exponentially small information is used, and the constants in L and b are suprema of explicit expressions in the inner solution (`proof/B5_inner.md`, §§2–4). For $M=1$, with $\varpi=\vartheta_0=0.3$ and $k_1=0.6$, every constant is certified in ball arithmetic over a complex parameter box containing the box of Theorem A($M=1$), and for data $\delta\le(1,200,10,20)$ the fixed point holds for $n\ge e^{73.89}$ with $x=(392,9.2\times10^4,21,30)$ (computer-assisted, `solver/proof/b5_bounds.py`). The threshold is set by the radius $|z|\approx0.009$ on which a perturbative fixed point around the inner series closes: the outer edge $k_1/\ln n$ of $\mathcal{M}^\pm$ must lie inside it.

Part (2) is proved given part (1). Both jumps are fast-type: $\Delta_{\rm in}=Z_{\rm in}^+-Z_{\rm in}^-=e^{-1/(2z)}z^{\langle F^2\rangle}e^{\Pi}k_{\rm in}(\theta_0)(1+O(z))$ with $k_{\rm in}=S$, and $\Delta_{\rm out}$ has the same form up to the factor (6.2). At the evaluation point $z_*=k_1/\ln n$ on $\arg z=0$, $|e^{-1/(2z_*)}|=n^{-1/(2k_1)}$, small only polynomially, so

$$k_{\rm out}-k_{\rm in}=\frac{(W^+-W^-)(z_*)}{e^{-1/(2z_*)}z_*^{\langle F^2\rangle}}+k\cdot O\Big(\frac{(\ln n)^2}{n}\Big)=O\big(n^{-1+1/(2k_1)}(\ln n)^{\mathrm{Re}\langle F^2\rangle}\big)$$

by part (1), which is $o(1)$ since $k_1>\frac12$; translating to outer label harmonics multiplies by $\kappa_{n,\mu}$. Replacing $Z_{\rm in}$ by $Z_{\rm in}+n^{-1}Z^{(1)}$, with $Z^{(1)}$ the first inner correction forced by $\mathcal{Q}_1$, would give the bracket $S_\mu+n^{-1}S_\mu^{(1)}$ plus a smaller error, i.e. the rate $O(1/n)$ of Theorem B(M); part (1) has not been run in that form.

Lemma B8(M) (global continuation). *For each $K_0\ge K_*$ and $n\ge n_0$ the centre-regular solution continues analytically to $\mathcal{D}_n^\pm$, and there*

$$Y^\pm(x)=\sum_{k\le K_0}\varepsilon^k\,Y_k(x)+R^\pm(x),\qquad |R^\pm(x)|\le C_{K_0}\,(1+|x|)\Big(\frac{(1+|x|)^2}{n}\Big)^{K_0+1},$$

with Y_k the polynomial coefficients of Proposition 1, of degree at most $2k+1$. For $n\ge n_0$ it supplies hypothesis (H8) of Lemma B5(1).

On $\mathcal{D}_n^\pm$, $(1+|x|)^2/n\le2k_1/\ln n$, so $Y^\pm$ has no exponentially large fast content, as Lemma B5(1) needs. The proof (`proof/B8_constants.md`) is a bootstrap in weighted norms of this form: the fast components ($p+m$ per harmonic, and $b-1$) are integrated along the paths of Lemma B8.1, gaining $1/(n|\Phi'(x)||x|)\sim|x|^2/n\le k_1/\ln n$ at large $|x|$ and $O(1/n)$ on bounded sets; the slow components are integrated from $x=0$, where regularity fixes them; the transport factor is absorbed in the fast-kernel bound, so it puts no condition on K_0; and the light-cone disc is never entered. It is proved for every fixed M, without a numerical $n_0(M)$. For $M=1$ the lemma holds

with explicit constants for $n \geq e^{62}$ ($K_0 = 4$, on the domain with R_n replaced by $(k_1 n/(\cos\vartheta_0 \ln n))^{1/2}$, in the gauge $c(0)$ equal to its formal value), and it supplies (H8) for $n \geq e^{73.89}$ with $\delta = (8.3\times10^{-5}, 79.04, 3.232, 6.464)$ (computer-assisted, in ball arithmetic). No Gevrey bound uniform up to $|x| \sim R_n$ is needed, because the exponentially small information enters only at the evaluation point z_*.

6.5. **Status of Theorem B(1).** For $M = 1$ the model is that of Definition 6.1, which is also the model of Theorem A($M = 1$), and every link of the chain is proved or certified except the following, so Theorem B(M) for $M = 1$ is not yet proved (`proof/C_selection.md`, §13.1, lists every link with its status and its threshold in n):

- the error in Lemma B5(2) is only estimated: for the implicit function theorem on the box of Theorem A($M = 1$) it must be about 10^{-7} or less, including its derivative, which needs $\ln n \gtrsim 285$ at $k_1 = 0.6$, where Lemmas B5(1) and B8 are certified, or $\ln n \gtrsim 104.7$ at $k_1 = 0.85$, where Lemma B8 has not been run;
- the block-diagonalization in Lemma B6(M) is not written;
- the error control of the outer transport of the jump along $x > 1$, from the light cone to the inner arc, which enters the explicit factor $\kappa_{n,\mu}$, is not written (Lemmas B6′ and B6 cover $x < 1$);
- Lemma B8 is certified in the gauge $c(0) = c^K(0)$, the light-cone gauge up to $O(n^{-5})$; its extension to the exact light-cone gauge is proved by hand but has not been run;
- the certificate code of Lemmas B5(1) and B8 has not been reviewed independently.

Braaksma's theorem is assumed, as in Theorem A($M = 1$).

6.6. **Gap (b): the harmonic tail.** Every constant above depends on M. The selection theorem for the full problem needs them uniform in M, or a formulation in a space of functions analytic in a θ-strip, as in Costin and Tanveer's Borel summability for PDEs [17]. The repository's analysis of this gap (`proof/B_tail.md`) finds it real but not fatal, and narrower than its statement in Section 4:

- *The weights cancel in the tuned problem.* Stated without harmonics, the condition is $S \equiv 0$ on the real label circle, and it involves F only on the real line. The weights $e^{\pi|\nu|\omega/2}$ are the linear connection coefficients of each label harmonic: they multiply zero in $S \equiv 0$, and they cancel in each tuning equation, because the free and the forced parts of S_ν are jumps of the same mode, so that $f_\nu = -(\text{forced jump})/(2\times \text{linear coefficient})$ is an $O(1)$ functional of the lower harmonics. In the amplitude expansion of Section 8.1 the leading tuning ratio $q_\nu = f_{\nu+2}/f_\nu$ is independent of ν to six digits, $q_\infty = 0.008524 + 0.000808\mathrm{i}$ ($|q|A^2 = 0.0633$), with no trace of the factor $e^{2\cdot6.004} = 1.6\times10^5$; the tuned profile decays by 0.0574 per pair of harmonics, i.e. it is analytic on $|\operatorname{Im}\theta| < \sigma_F = 1.428$; and the neighbour-family amplitudes in the late orders fall by 10^4–10^8 per pair of harmonics instead of growing by about 25.
- *Naive truncation brings the weights back.* Projecting the amplitude recursion onto $|\mu| \leq M$ changes σ_M by about 5×10^{-4}, as order counting predicts, but gives σ_{M+1} too large by a factor 10^4–10^6 (-399 at $M = 3$, -1371 at $M = 5$). A tail lemma must therefore be stated for the *jump-conditioned* truncation ($S_3 = \cdots = S_M = 0$ imposed and the higher labels eliminated by a Schur complement), whose sequence converges (Section 9.2), not for a Galerkin projection of the equations.
- *The tail size looks provable.* The jump-conditioned sequence alternates and shrinks by a ratio of -0.045 to -0.057 per pair, the profile's own decay times a coupling of 0.7–1.0. That puts the tail above $M = 5$ at about 4–6 in the harmonic-1 residual r_1, above $M = 7$ at 0.2–0.35, and above $M = 9$ at about 0.015; a certified box of radius 10^{-3} needs the tail below about 4, so with $M = 7$ explicit the margin is about 13. Even a proven ratio of 0.25 per pair would suffice.
- *A framework.* For $0 < \varsigma < \sigma_F$ let X_ς be the odd trigonometric series with $\|F\|_\varsigma = \sum_\nu |F_\nu| e^{\varsigma|\nu|} < \infty$, an algebra of functions analytic on $|\operatorname{Im}\theta| < \varsigma$, with the multipliers measured in the dual weights. At leading order the tuning map is the weighted shift $f_{\nu+2} = q_\infty A^2 f_\nu$, of norm $0.0633\, e^{2\varsigma}$ on X_ς, which is below 1 for $\varsigma < 1.380$ (e.g. 0.47 at $\varsigma = 1$). The target is then: *in X_ς with $\varsigma \approx 1$, $\|(I - P_M)F\|_\varsigma \leq C_1(|q_\infty|A^2 e^{2\varsigma})^{(M+1)/2}$, and the residual of the jump-conditioned M-harmonic problem differs from the untruncated one by at most $L\,\|(I - P_M)F\|_\varsigma$, with explicit C_1, L such that the bound at $M = 7$ is below about 2.* The first half is a contraction estimate; the second, with L estimated for the tuned system, is the genuinely new part.
- *Secondary Stokes lines.* The singularities of F at $\operatorname{Im}\theta = \pm1.428$ reach complex ξ along $\arg\xi \to \mp0.374$, between the Stokes and anti-Stokes lines. They are absent at each finite order in the amplitude (every

order is a trigonometric polynomial); where they sit at $|\xi| = O(1)$ has not been computed, so a proof should define S_1 order by order in A^2 and sum in A^2 last.

7. Theorem A($M = 1$): a computer-assisted proof

This section proves that the selection condition of the one-harmonic truncation of the inner problem has exactly one zero in an explicit box, and that it is simple. The model is a test case, not the physical problem: its zero lies 22% below ω_∞ in frequency and does not survive the addition of harmonic 3 (Section 9.2). Its role is to show that the inner multipliers can be certified, along a route that extends to more harmonics. The details are in `proof/A_certify.md` §§4–12 and the independent review `proof/A_review.md`; the code is in `solver/proof/`. Notation in this section is local.

7.1. The one-harmonic problem.

Let $\mathbf{x} = (A, \omega/\omega_\infty)$ with $\omega_\infty := 2\pi/1.64389375$ (a convention, Section 2.2), and $F = A\sin\theta$, so $F_{\pm1} = \mp \mathrm{i}A/2$. The $M = 1$ truncation of (4.2) keeps y and σ on the harmonics $\mu = \pm1$ and c on $\nu = 0, \pm2$, and projects every product onto its result's harmonics, written $[\cdot]_\mu$:

$$\xi^2 y'_\mu = [c\sigma]_\mu - \mathrm{i}\mu\omega\xi\, y_\mu, \qquad \xi^2\sigma'_\mu = [c\,\xi y']_\mu + [c\,e\,y]_\mu - [c\,y]_\mu - \mathrm{i}\mu\omega\xi\,\sigma_\mu, \qquad \xi\, c'_\nu = [c\,e]_\nu, \tag{7.1}$$

with $e = \frac{1}{2}[yy]$ and $' = \mathrm{d}/\mathrm{d}\xi$. This is a system for seven complex unknowns, polynomial in them and rational in ξ, with an irregular singular point at $\xi = 0$. Its formal solution $y_\mu = \sum_j d_{j,\mu}\xi^{2j+1}$, $\sigma_\mu = \sum_j s_{j,\mu}\xi^{2j+2}$, $c_\nu = \sum_j g_{j,\nu}\xi^{2j}$ with $d_{0,\mu} = 2F_\mu$, $c = 1 + O(\xi^2)$ is unique and Gevrey-1 in $z = \xi^2$. We write $\mathsf{y} = (y_{\pm1}, \sigma_{\pm1}, c_{0,\pm2})$ for the state.

Definition 7.1 (lateral solutions and the selection condition)**.** *Let* $\mathbf{x}$ *be real. The* lateral solution y^+ *is the solution asymptotic to the formal series on a sector of small radius containing* $0 \le \arg z \le 3\pi/4$*; its existence and uniqueness are given by Braaksma's theorem* [7] *(Lemma B4(M)),* the one fact taken from the literature. *The reflection* $\mathcal{R}(\mathsf{y})_\mu = \overline{\mathsf{y}_{-\mu}}$ *(on each of* y*,* σ*,* c*) maps solutions to solutions under* $\xi \mapsto \bar\xi$ *and fixes the formal series, so* $\mathsf{y}^-(\xi) = \mathcal{R}(\mathsf{y}^+(\bar\xi))$ *is the lateral solution from below. The* $M = 1$ selection condition *is* $\mathsf{y}^+ \equiv \mathsf{y}^-$*. With* $\mathsf{y}^\pm_{\mathbb{R}}$ *the continuations along the real axis, put*

$$\psi = \mathsf{y}^+_{\mathbb{R}} - \mathsf{y}^-_{\mathbb{R}} = \mathsf{y}^+_{\mathbb{R}} - \mathcal{R}(\mathsf{y}^+_{\mathbb{R}}), \qquad J(\xi_1; \mathbf{x}) = \psi_{y,+1}(\xi_1) = y^+_{+1}(\xi_1) - \overline{y^+_{-1}(\xi_1)},$$

so that $\psi_{y,-1}(\xi_1) = -\overline{J(\xi_1)}$ *and* $\mathcal{R}\psi = -\psi$*. (*J *is* J_D *in* `proof/A_certify.md`*.) The single complex Stokes constant* $\Sigma(\mathbf{x})$ *is defined in* (7.7) *below.*

In the linear problem the jump is known in closed form (Section 4), and in ball arithmetic it equals $2\pi\mathrm{i}K'(s)\,2F_1$ times the recessive solution to $1 \pm 3 \times 10^{-30}$. The computation follows a path: it continues y^+ from a start point $\xi_0 = 0.1\,e^{3\mathrm{i}\pi/8}$ on the ray $\arg\xi = 3\pi/8$, along that ray and then down to the real axis. Define $J^{\mathrm{path}}(0.3; \mathbf{x})$ as J computed with y^+ continued along the *lower path*: the ray from 0.1 to 0.2, the arc $|\xi| = 0.2$ down to the real axis, and the real axis from 0.2 to 0.3.

7.2. The template and the plan of the proof.

The template is the computer-assisted proof of a Stokes constant by Baldomá, Capiński, Guardia and Seara [4]: two solutions characterized by their asymptotics on two sectorial domains; a fixed point for each in a weighted space, with explicit bounds on the forcing and on a Lipschitz constant; validated values far out; and the Stokes constant from the difference of the two solutions, by a contraction for that difference. We need a *zero* of the multiplier in two real parameters, not a non-zero value, so an interval-Newton (Krawczyk) step is added. The proof has these parts:

(i) a start-value fixed point encloses y^+ near $\xi = 0$, with parameter derivatives (Lemma 7.2);
(ii) a validated Lohner integration carries the enclosure along the path to $\xi_1 = 0.3$;
(iii) J comes from the same enclosure, using the reflection;
(iv) a Krawczyk test shows that $J^{\mathrm{path}}(0.3; \cdot)$ has exactly one zero in the box, and that it is simple;

together with two lemmas: *6b*, that the path value is the value of the real-axis continuation (analyticity of y^+ on the region the path sweeps; Lemmas S, 6b$_0$, 6b$_1$), and *6a*, that the one component J at the one point 0.3 vanishes only if the whole jump does (Lemma 6a). Everything is computed in ball arithmetic (the Arb library through python-flint 0.9, 128–320 bits); every input decimal (the centre of the box, its half-width, 0.1, 0.3, 1.64389375) enters as a decimal string, so every ball contains the exact decimal.

7.3. **Step (i): the start-value fixed point.** Let $z = \xi^2$ and, on each harmonic, $y_\mu = \xi P_\mu$, $\sigma_\mu = zQ_\mu$, $V_\mu = Q_\mu - (1+s_\mu)P_\mu$ with $s_\mu = \mathrm{i}\mu\omega$; let $W = V + [(C-1)Q]$ and $\hat{E} = [P^2/2]$, where C denotes c. For $z \neq 0$, (7.1) is equivalent to

$$P' = \frac{V + [(C-1)Q]}{2z}, \qquad C' = \tfrac{1}{2}\,[C\hat{E}],$$
$$V' = \frac{V}{2z^2} + \frac{[(C-1)Q] + [(C-1)W]}{2z^2} + \frac{-(2+s)\,Q - (1+s)\,W + [C\hat{E}P]}{2z}, \tag{7.2}$$

with $' = \mathrm{d}/\mathrm{d}z$. The only rank-one singular term is $V/(2z^2)$, whose solution $e^{-1/(2z)}$ is the fast mode; the slow fundamental solutions are constants, and every other term is a perturbation, since $C - 1 = O(z)$.

Write $\mathsf{Y} = \mathsf{Y}_N + \eta$, with Y_N the formal series truncated at $N = 53$ terms and $\eta = (\eta_P, \eta_V, \eta_C)$, and let $\Phi_\bullet$ be the right-hand side minus the principal term $V/(2z^2)$, evaluated at $\mathsf{Y}_N + \eta$, minus Y_N'; thus $\Phi = \rho_N +$ (the change of the right-hand side), with ρ_N the residual of the truncated series. Integrating from $z = 0$ along a ray,

$$\eta_P(z) = \int_0^z \Phi_P\,\mathrm{d}t, \qquad \eta_V(z) = \int_0^z e^{\frac{1}{2t} - \frac{1}{2z}}\,\Phi_V(t)\,\mathrm{d}t, \qquad \eta_C(z) = \int_0^z \Phi_C\,\mathrm{d}t. \tag{7.3}$$

On the ray $\arg z = \vartheta$ the fast kernel has modulus $e^{-a(1/|t| - 1/|z|)} \le 1$ for $|t| \le |z|$, with $a = -\frac{1}{2}\cos\vartheta = \sqrt{2}/4$ at $\vartheta = \pm 3\pi/4$, and with $r = |z|$

$$\int_0^r \varrho^m\,\mathrm{d}\varrho = \frac{r^{m+1}}{m+1} \quad (m > -1), \qquad \int_0^r e^{-a\left(\frac{1}{\varrho} - \frac{1}{r}\right)}\varrho^m\,\mathrm{d}\varrho \le \frac{r^{m+2}}{a} \quad (m \ge -2), \tag{7.4}$$

the second by the substitution $1/\varrho$ and $\varrho^m \le r^{m+2}\varrho^{-2}$. On a ray at angle ϑ the rate a stays positive for $\pi/2 < \vartheta < 3\pi/2$; on $|\arg\xi \mp 3\pi/8| \le 0.2$ its smallest value is 0.188.

Let $R = 0.01$ and $k = 40$, and use on $0 < r \le R$ the norms $\sup r^{-k}|\eta_P|$, $\sup r^{-k}|\eta_V|$ and $\sup r^{-k-1}|\eta_C|$ for each of the seven components, combined into a weighted maximum with weights from the Perron vector of the Lipschitz matrix (any positive weights are valid). Every function is bounded by a majorant $K|t|^q$; products and harmonic convolutions are bounded by the triangle inequality, derivatives by the product rule on majorants, and the Lipschitz matrix by the mean-value inequality for holomorphic maps on the convex ball. Every bound is an upper bound in ball arithmetic and non-decreasing in R.

Lemma 7.2 (start values)**.** *Let* $\mathbf{x}$ *be the point* $\hat{\mathbf{x}} = (2.5421480475, 0.7769940136)$ *or any point of the complex box around it of half-width* 3×10^{-2} *in the real and in the imaginary part of each parameter. On each ray* $\arg\xi = \pm 3\pi/8$ *with* $0 < |\xi| \le 0.1$*, the truncated system has a solution* Y *with* $\mathsf{Y} - \mathsf{Y}_N$ *in an explicit ball of the weighted space, and it is the only solution in that ball; every solution with* $\mathsf{Y} - \mathsf{Y}_N = o(|\xi|^{82})$ *along the ray is this one, and* y^+ *(on the upper ray) and* y^- *(on the lower ray) are such solutions. At* $\xi_0 = 0.1\,e^{3\mathrm{i}\pi/8}$ *and* $\hat{\mathbf{x}}$*, the enclosure has radius* 2.4×10^{-34} *in* y*,* 6.0×10^{-34} *in* σ *and* 5.9×10^{-36} *in* c *on every harmonic. The same holds on every ray with* $|\arg\xi \mp 3\pi/8| \le 0.2$*.*

Proof. Forcing. The numerators of ρ_N are polynomials in z whose coefficients below order 53 (for P and V) and 52 (for C) vanish identically, because the recursion solves the order-by-order equations exactly; the computation checks that each of these balls contains 0, and the independent review verified the vanishing exactly in $\mathrm{GF}(p)$ at random parameters (the chance of a false pass is below 10^{-50}). The rest is bounded and pushed through (7.4), which gives $F = \|T(0)\|$ for the operator T of (7.3). For $|\xi| \le 0.1$ the residual itself is at most 1.0×10^{-33} in every direction.

Contraction. With Λ the Lipschitz constant of T on the ball of radius ϱ, the check is $\Lambda < 1$ and $F + \Lambda\varrho < \varrho$:

case	$\Lambda \le$	F	ϱ	$F + \Lambda\varrho$
$\hat{\mathbf{x}}$, rays $\arg\xi = \pm 3\pi/8$	0.2918	1.32×10^{48}	7.43×10^{48}	3.48×10^{48}
complex box $\pm 10^{-3}(1+\mathrm{i})$	0.2923	8.57×10^{69}	4.84×10^{70}	2.27×10^{70}
complex box $\pm 3 \times 10^{-2}(1+\mathrm{i})$	0.3048	1.17×10^{72}	6.73×10^{72}	3.22×10^{72}
$\hat{\mathbf{x}}$, all rays $\lvert\arg\xi \mp 3\pi/8\rvert \le 0.2$	0.5107	1.32×10^{48}	1.08×10^{49}	6.80×10^{48}

(F and ϱ are in the weighted norm and carry the factor $R^{-k} = 10^{80}$; in absolute terms the ball at $\hat{\mathbf{x}}$ is $|\eta| \le 5.3 \times 10^{-32}$.) So T maps the ball into itself and contracts.

Identification. Every bound is non-decreasing in R, so the contraction holds on each $(0, R']$, $R' \le R$, and, the operator being of Volterra type, the fixed point on $(0, R]$ restricts to the fixed point on $(0, R']$. A solution with $\eta_P, \eta_V = o(r^k)$ and $\eta_C = o(r^{k+1})$ satisfies (7.3); the boundary terms vanish because $e^{1/(2\epsilon)} \to 0$ as $\epsilon \to 0$

along the ray, where the fast mode is exponentially large. Its weighted norm on $(0, R']$ tends to 0, so for small R' it lies in the ball and is the fixed point there, and by uniqueness for the regular ODE it is the fixed point on all of $(0, R]$. Finally $\mathsf{y}^+ - \mathsf{Y}_N = O(z^{53})$ on the upper ray by Definition 7.1, and $53 > k+1$; the lower ray gives y^- by the reflection. □

Parameter derivatives. The Krawczyk test needs $\partial \mathsf{y}/\partial \mathbf{x}$ at ξ_0 over a real box X. The derivatives of Y_N are exact (forward-mode jets [47] through the recursion). Since the contraction holds uniformly on the complex box $X + c_0(1+\mathrm{i})$, $c_0 = 10^{-3}$, η is holomorphic in the parameters there, and Cauchy's estimate gives $|\partial_{x_i}\eta| \le c_0^{-1} \sup_{X+c_0(1+\mathrm{i})} |\eta|$; at ξ_0 this adds less than 10^{-8} to the derivative enclosures.

7.4. **Step (ii): validated integration.** The integrator (`validated_int.py`) is a QR-Lohner method [31, 35] in ball arithmetic at 256 bits, with ω and A carried as constant states and the set represented in real coordinates as $c + Qq$ (c a point, Q a real 18×18 matrix, q a box); every step is a multiple of 2^{-30} of the leg parameter, so every node is exact. One step consists of an a-priori enclosure of all trajectories from the set by interval Picard iteration; Taylor polynomials of order 24 of the centre trajectory and of the variational matrix; remainders from the order-25 coefficients on the a-priori enclosure; and the mean-value update $\phi(c+Qq) \subset \phi(c) + JQq$, re-orthogonalized by the Q factor of $\mathrm{mid}(JQ)$, whose inverse is enclosed rigorously. The stiff fast modes therefore only shrink q, whereas a naive interval Taylor method would overestimate by about e^{50} on the ray. The columns are pivoted by their norm times the size of the matching component of q (Lohner's rule); with norms alone the enclosure grew by about 15% per step at the start of the arc, where the flow is strongly non-normal, against 2–3%. For the derivatives over a box (the *box run*) each column $\partial \mathsf{y}/\partial x_k$ is a Lohner set of its own, started from the data of step (i), and the state set over the box is rebuilt at every step as $\mathsf{y}(t;\hat{\mathbf{x}}) + V(t;X)(\mathbf{x} - \hat{\mathbf{x}})$, with V the derivative enclosure.

On the path ray–arc at radius 0.3 (743 steps), the point run gives $J(\hat{\mathbf{x}}) = 1.054740 \times 10^{-12} - 3.490429 \times 10^{-12}\mathrm{i}$ with radius 4.1×10^{-19}, in agreement with a double-precision solver to 9×10^{-16}; on the lower path (303, 322 and 47 steps on the three legs) the same $J(\hat{\mathbf{x}})$ to the digits shown, with radius 6×10^{-17}. The box runs are the bottleneck: the excess of the derivative enclosures grows like $7 \times 10^5\, h$ in the half-width h of the box, because the solution is large where the ray meets the arc, and they diverge for $h \ge 10^{-4}$.

7.5. **Step (iv): the Krawczyk test.** Let $X = \hat{\mathbf{x}} + [-h, h]^2$ and $g(\mathbf{x}) = \kappa_0\, J^{\mathrm{path}}(0.3; \mathbf{x})$, read as a map $\mathbb{R}^2 \to \mathbb{R}^2$, where $\kappa_0 = \xi_1^{-\hat{A}^2}/N(\hat{\mathbf{x}})$ is a non-zero constant enclosed rigorously, N being the linear normalization $2\pi\mathrm{i}K'(s)\,2F_1$ times the recessive solution at ξ_1. Let $J(X)$ enclose $\{Dg(\mathbf{x}) : \mathbf{x} \in X\}$ (from the box run; for real $\mathbf{x}$ the derivative of the conjugate is the conjugate of the derivative), and let Y be any real 2×2 matrix (the code takes a floating-point inverse of $\mathrm{mid}\, J(X)$). Put

$$K(X) = \hat{\mathbf{x}} - Y\, g(\hat{\mathbf{x}}) + \big(I - Y\, J(X)\big)(X - \hat{\mathbf{x}}). \tag{7.5}$$

If $K(X) \subset \mathrm{int}\, X$, then g has exactly one zero in X and Dg is non-singular there (Krawczyk [28], Moore [34], Neumaier [36]). The inputs are g at the single point $\hat{\mathbf{x}}$ and J over the whole box.

Results. With $h = 10^{-9}$ the test passes on the ray–arc path at radius 0.3, with margins 7.9×10^{-10} and 9.2×10^{-10} (out of 10^{-9}); there $J(X)$ has midpoints -15.32, -103.12, 90.86, -44.09 and radii 2.7–3.5, about a tenth of what the test tolerates. On the lower path it passes with margins 8.570×10^{-10} and 9.712×10^{-10}; since $J(X)$ also encloses the Jacobian over every sub-box centred at $\hat{\mathbf{x}}$, it passes on $\hat{\mathbf{x}} \pm 1.2 \times 10^{-10}$ as well, which puts the zero of $J^{\mathrm{path}}(0.3; \cdot)$ in

$$A \in [2.54214804761338,\ 2.54214804761975], \qquad \omega/\omega_\infty \in [0.776994013607438,\ 0.776994013612010] \tag{7.6}$$

(endpoints rounded outward; $\omega \approx 2.96977670855$). Larger boxes fail because of the derivative excess; a C^2 Lohner step would be needed for 10^{-3} boxes.

Mutation controls. A test that has never been seen to fail is untested, so the function (never an enclosure) was mutated and the full assembly rerun from the saved runs, against verdicts derived from the construction (`mutation_controls.py`):

	mutation	expected	result
C0	none	pass	pass, margins 7.925×10^{-10}, 9.247×10^{-10}
C1	jump sign flipped, $y^+_{+1}+\overline{y^+_{-1}}$	fail	fail, margins -59, -22
C2	conjugation dropped, $y^+_{+1}-y^+_{-1}$	fail	fail, margins -2.0, -0.35
C3	equivalent form $-\overline{J}$	pass, same zero	pass; K overlaps C0's
C4	normalization $N\to 2N$	pass, same K	pass; K identical to 30 digits
C5	zero planted outside the box, moved by $(2h,0)$	fail	fail, margin -1.2×10^{-9} in A
C6	zero planted inside, moved by $(h/2,-3h/10)$	pass, zero recovered	pass; K contains the moved zero
C7	point enclosure of $y^+_{+1}(\xi_1)$ widened by 10^{-9}	fail	fail, margins -4×10^{-8}

A planted zero is made by subtracting Ld from g, with L the exact midpoint of $J(X)$, which moves the zero by d up to second order and leaves $J(X)$ unchanged. The controls were designed after the certificate had passed, which weakens them somewhat; their expected verdicts follow from the construction, not from the real run.

7.6. **Lemma S and the analyticity near the origin.** The rays of Lemma 7.2 stay at $|\arg z-3\pi/4|\le 0.4$. To reach the real axis, change the integration path of η_V and nothing else. Let $R_+=0.01$, $U_0=\sqrt{2}/R_+$, $\mathcal{K}=\{w:-3\pi/4\le\arg w\le 0\}$, and

$$\mathcal{D}_S=\{z:1/z\in U_0+\mathcal{K}\}.$$

Elementary geometry of the translated cone gives: $|z|\le R_+$ on $\mathcal{D}_S$; in the direction $\arg z=\vartheta\in[0,3\pi/4)$, $\mathcal{D}_S$ contains every z with $|z|\le R_+\sin(3\pi/4-\vartheta)$, which is at least $R_+/\sqrt{2}$ for $\vartheta\le\pi/2$ and includes the real segment $0<z\le R_+/\sqrt{2}$; $\mathcal{D}_S$ is star-shaped about 0, so η_P and η_C are still integrated along $[0,z]$; and for $u\in U_0+\mathcal{K}$ the path $u+s'e^{-3\mathrm{i}\pi/4}$, $s'\ge 0$, stays in $U_0+\mathcal{K}$ and ends at $z\to 0$ along $\arg z\to 3\pi/4$. For $z\in\mathcal{D}_S$ integrate the V equation along that path: $\eta_V(z)=\int_{\gamma_z}e^{1/(2t)-1/(2z)}\Phi_V(t)\,\mathrm{d}t$ with $1/t=1/z+s'e^{-3\mathrm{i}\pi/4}$, s' from ∞ to 0. Along it the kernel has modulus $e^{-as'}$, $a=\sqrt{2}/4$, and $|\mathrm{d}t|=|t|^2\mathrm{d}s'$. With $u=1/z$, $c_\phi=-\cos\phi$ (ϕ the angle between u and $e^{-3\mathrm{i}\pi/4}$) and $X'=s'/|u|$,

$$\frac{|u|^2}{|u+s'e^{-3\mathrm{i}\pi/4}|^2}=\frac{1}{1-2c_\phi X'+X'^2}\le e^{2c_\phi X'}\qquad(0\le c_\phi\le 1/\sqrt{2}),$$

because $f(X')=\ln(1-2c_\phi X'+X'^2)+2c_\phi X'$ has $f(0)=0$ and $f'=2X'(1-2c_\phi^2+c_\phi X')/(1-2c_\phi X'+X'^2)\ge 0$; for $c_\phi\le 0$ the ratio is at most 1. On $\mathcal{D}_S$, $c_\phi\le 1/\sqrt{2}$ always and $c_\phi/|u|\le 1/(\sqrt{2}U_0)=R_+/2$. Hence

$$\int_{\gamma_z}\left|e^{\frac{1}{2t}-\frac{1}{2z}}\right||t|^m\,|\mathrm{d}t|\le|z|^{m+2}\int_0^\infty e^{-as'+(m+2)\frac{c_\phi}{|u|}s'}\,\mathrm{d}s'\le\frac{|z|^{m+2}}{a-(m+2)R_+/2},$$

and in the weighted norm the fast bound takes the form of (7.4) with a replaced by $a_{\mathrm{eff}}=\sqrt{2}/4-kR_+/2$. The Perron weights are made symmetric under $\mu\mapsto-\mu$, so that $\mathcal{R}$ maps the ball to itself.

Lemma S. *Let $X=\hat{\mathbf{x}}+[-10^{-9},10^{-9}]^2$. For every $\mathbf{x}\in X$ the truncated system has a solution $\mathsf{Y}=\mathsf{Y}_N+\eta$ on $\mathcal{D}_S$, with η analytic in the interior of $\mathcal{D}_S$ and in an explicit weighted ball; it is unique in that ball, and it is y^+. (With $k=30$, $N=53$: $a_{\mathrm{eff}}=0.20355$, contraction constant at most 0.4903, forcing 1.04×10^{44}, ball 8.14×10^{44}, and $F+\Lambda\varrho=5.03\times10^{44}$.)*

Proof. The contraction is checked as in Lemma 7.2 (`lemma6_sector.py`). For the identification, the domains $\mathcal{D}_S(U_0')=\{1/z\in U_0'+\mathcal{K}\}$ decrease as U_0' grows, the operator on $\mathcal{D}_S(U_0')$ uses only values there, and every bound is non-decreasing in R_+, so the contraction holds on each $\mathcal{D}_S(U_0')$, $U_0'\ge U_0$. For large U_0', $\mathcal{D}_S(U_0')$ lies in the sector of asymptoticity of y^+, where $\mathsf{y}^+-\mathsf{Y}_N=O(|z|^{53})$; so $\mathsf{y}^+-\mathsf{Y}_N$ satisfies the integral equation there (the boundary terms of η_P, η_C vanish at 0, and that of η_V is $e^{-as'}$ times a bounded function), its weighted norm tends to 0, and it is the fixed point. Both are solutions of the ODE on the connected set $\mathcal{D}_S$, so they agree on $\mathcal{D}_S$. □

Lemma 6b$_0$. *For every $\mathbf{x}\in X$, y^+ extends analytically to the interior of $\Omega_0=\{0<|\xi|\le 0.0624,\ 0\le\arg\xi\le 3\pi/8\}$, continuously to its boundary, and to the real segment $0<\xi\le 0.0841$, where it is enclosed by Lemma S.*

Proof. $\mathcal{D}_S$ contains the directions $0\le\arg z\le 3\pi/4-0.4$ up to radius at least $R_+\sin 0.4=0.003894$, i.e. $|\xi|\le 0.0624$, and the real segment up to $R_+/\sqrt{2}$, i.e. $\xi\le 0.0841$. The sector version of Lemma 7.2, re-proved over X (contraction 0.5107), covers $3\pi/4-0.4\le\arg z\le 3\pi/4$ up to $|z|=0.01$; its fixed points on these rays form one analytic function, because the ray integrals from 0 are path-independent there ($\mathrm{Re}(1/t)\to-\infty$ uniformly as $t\to 0$ in that sector), and it is y^+. The two pieces overlap in a connected set on which both are y^+. □

7.7. **Lemma 6a: one component at one point is the whole jump.** Put $\xi_a = 0.067$ and $Z = \xi_a^2 = 0.004489 \le R_+/\sqrt{2}$. In the variables of (7.2), let $\beta = \eta_V$-component of ψ at $z = Z$, $\beta \in \mathbb{C}^2$; it satisfies $\beta_{-1} = -\overline{\beta_{+1}}$. Define

$$\Sigma(\mathbf{x}) := \beta_{+1} = (\text{fast component of the jump } \psi \text{ on harmonic } +1 \text{ at } z = Z), \tag{7.7}$$

the single complex Stokes constant of the $M = 1$ problem.

Lemma 6a. *Let B_1 be the state-set box at $\xi = 0.3$ of the lower-path box run. For every $\mathbf{x} \in X$: if $\mathsf{y}^+_{\mathbb{R}}(0.3; \mathbf{x}) \in B_1$, then*

$$J(0.3; \mathbf{x}) = 0 \iff \psi \equiv 0 \iff \mathsf{y}^+ \equiv \mathsf{y}^- \iff \Sigma(\mathbf{x}) = 0.$$

Proof. The implications from right to left are immediate except $\Sigma = 0 \Rightarrow \psi \equiv 0$, which is step 1.

Step 1 (flat estimate on $0 < z \le Z$). By Lemma S, $\mathsf{y}^+ = \mathsf{Y}_N + \eta^+$ there with η^+ in the ball; by the reflection and the symmetric weights $\mathsf{y}^- = \mathsf{Y}_N + \mathcal{R}(\eta^+)$ is in the same ball, and so is the segment between them. So $\psi = \eta^+ - \eta^-$ satisfies exactly $\psi = b + T_0\psi$, with $b = (0, e^{1/(2Z)-1/(2z)}\beta, 0)$, where T_0 integrates the difference of the perturbation terms (linear in ψ by the mean-value theorem on the ball): the slow components from 0 (valid because $\psi = O(z^k)$), and the fast one backward from Z by variation of constants. The backward kernel obeys, for $m + 2 \ge 0$ and $(m+2)Z < \frac{1}{2}$,

$$\int_z^Z e^{-\frac{1}{2}\left(\frac{1}{z}-\frac{1}{t}\right)} t^m \, \mathrm{d}t \le \frac{z^{m+2}}{\frac{1}{2} - (m+2)Z}$$

(substitute $1/t$ and use $(U/u)^{m+2} \le e^{(m+2)Z(U-u)}$), so the weighted fast bound has $a_{\text{eff}} = \frac{1}{2} - kZ = 0.3653$, and with the derivative majorants over the ball of Lemma S, $\|T_0\| \le \Lambda = 0.1513$. Since $z^{-k}e^{-1/(2z)}$ increases on $(0, Z]$, $\|b\| = Z^{-k} \max_\mu \mathsf{w}_{V,\mu}|\beta_\mu|$, with w the weights. Hence $\epsilon := \|\psi - b\| \le \Lambda(1-\Lambda)^{-1}\|b\|$, and $|\psi_{P,\mu}(Z)| \le Z^k\epsilon/\mathsf{w}_{P,\mu}$, $|\psi_{C,\nu}(Z)| \le Z^{k+1}\epsilon/\mathsf{w}_{C,\nu}$, while the fast component at Z is β. In particular $\beta = 0$ gives $\psi = T_0\psi$, hence $\psi = 0$ on $(0, Z]$.

Step 2 (the backward flow). If $J(0.3) = 0$, then $\psi(0.3) = (0, 0, \mathbf{u})$, the y-components vanishing, with $\mathbf{u} = (\psi_\sigma, \psi_c) \in \mathbb{C}^5$. The C^1 Lohner integrator runs backward along the real axis from 0.3 to ξ_a, where the fast modes decay by about e^{-106}, from the hull of B_1 and $\mathcal{R}(B_1)$ with the identity as variational columns (291 steps; largest radius 1.8×10^{-6}). The resulting enclosure B contains the flow Jacobian at every point of that hull, so by the mean-value theorem $\psi(\xi_a) \in B\,\psi(0.3)$. In the variables of (7.2) at ξ_a, with the exact change of variables T_a and $G = T_aB$ restricted to the columns of $\mathbf{u}$, $(\psi_P, \psi_C)(Z) = G_{\text{slow}}\mathbf{u}$ and $\beta = G_{\text{fast}}\mathbf{u}$.

Step 3 (contraction in β). G_{slow} (5×5) is invertible: the ball is inverted rigorously, with entries of G_{slow}^{-1} at most 1.26. So $\beta = H\,(\psi_P, \psi_C)(Z)$ with $H = G_{\text{fast}}G_{\text{slow}}^{-1}$, whose entries are at most 7.7. With step 1,

$$\max_\mu \mathsf{w}_{V,\mu}|\beta_\mu| \le \rho_{6a} \max_\mu \mathsf{w}_{V,\mu}|\beta_\mu|, \qquad \rho_{6a} = \frac{\Lambda}{1-\Lambda} \max_\mu \mathsf{w}_{V,\mu} \sum_j |H_{\mu j}| \frac{Z^{k_j - k}}{\mathsf{w}_j} \le 4.22 \times 10^{-3}.$$

So $\beta = 0$, and $\psi = 0$ on $(0, Z]$ by step 1. By uniqueness for the ODE, $\mathsf{y}^+_{\mathbb{R}} \equiv \mathsf{y}^-_{\mathbb{R}}$, and the two germs coincide. □

The margin is large ($\rho_{6a} = 0.004$ against 1). A relative change of 10^{-6} in one series coefficient makes Lemma S fail, and dropping the relabelling $\mu \mapsto -\mu$ in the reflection makes the backward integration fail; but $(1 - \mathrm{i}\mu\omega)$ in place of $(1 + \mathrm{i}\mu\omega)$ in the change of variables at ξ_a is not caught ($\rho_{6a} = 0.12$), since the wrong V differs by $2\mathrm{i}\mu\omega P$ and $P(Z)$ is itself small. That formula is guarded instead by self-tests against the integrator's system.

7.8. **Lemma 6b$_1$: a covering by validated Taylor discs.** It remains to show that y^+ is analytic on the annular sector between $|\xi| = 0.0624$ and 0.2. A first-order family of paths (tubes swept with the offset as a state) blows up beyond widths of about 0.1–0.2% of the radius, because the parameter width enters every step Jacobian. Instead we use that a translated family $\Psi(t, d) = \mathsf{y}(u_c(t) + d)$ is the Taylor expansion of y around $u_c(t)$, whose natural radius is the distance to the nearest singularity of y^+. Work in $u = 1/\xi^2$, where the fast mode is $e^{-u/2}$ with a constant rate, with the state $\mathsf{Y} = (P_{-1}, P_{+1}, V_{-1}, V_{+1}, 100C_{-2}, 100(C_0 - 1), 100C_{+2})$; the system $\mathsf{Y}' = \mathcal{F}(u, \mathsf{Y}; \mathbf{p})$ is polynomial in Y and in $1/u$, with parameters $\mathbf{p}$. In double precision the nearest singularity of y^+ is a branch point at $u^* = 9.6746\,e^{-2.1198\mathrm{i}}$ ($\xi^* = 0.3215\,e^{1.0599\mathrm{i}}$), at distance 15 or more from the annulus.

Lemma D (disc lemma). *Let c and $\mathbf{p}$ be complex balls, Y_0 a box in $\mathbb{C}^7$, $N_T \ge 1$ and $0 < r < |c|$, and suppose the computation below succeeds. Then for every exact $c' \in c$, $\mathsf{y}_0 \in \mathsf{Y}_0$ and $\mathbf{p}' \in \mathbf{p}$, the solution with $\mathsf{y}(c') = \mathsf{y}_0$ extends analytically to the open disc $|u - c'| < r$, continuously to its closure, and $|\mathsf{y}_j(u) - \mathsf{y}_{0,j}| \le M_j$ there.*

The computation: (1) the Taylor coefficients p_k, $k \le N_T$, of the solution in $\tau = u - c$, by Picard iteration on power series in ball arithmetic from the balls; they enclose the Taylor coefficients of every solution in the statement. (2) The defect $d = \mathcal{F}(c+\tau, p(\tau)) - p'(\tau)$ as a polynomial, with $1/u$ expanded as $\sum_{j\le J}(-\tau)^j/c^{j+1} + \tau^{J+1}\zeta(\tau)$, $|\zeta(\tau)| \le |c|^{-J-2}(1-r/|c|)^{-1}$, $J = N_T+8$, and ζ replaced by a ball of that radius. (3) The coefficients of d below order N_T vanish identically for each exact initial value; the code checks that their balls contain 0 and drops them, so $|d_i(\tau)| \le \Delta_i|\tau|^{N_T}$ with $\Delta_i = \sum_{k\ge N_T}|d_{i,k}|r^{k-N_T}$. (4) With $M_j^0 = \sum_{1\le k\le N_T}|p_{j,k}|r^k$, the box $B_j = \mathsf{Y}_{0,j} + \mathrm{disc}(M_j^0 + b_j)$, a ball containing $1/u$ on the disc, and $L_{ij} \ge |\partial\mathcal{F}_i/\partial \mathsf{y}_j|$ over all of these, form, with positive weights w,

$$\Lambda = \frac{r}{N_T+2}\max_i \sum_j \frac{\mathsf{w}_i L_{ij}}{\mathsf{w}_j}, \qquad F_0 = \max_i \frac{\mathsf{w}_i\Delta_i}{N_T+1}, \qquad \varrho = \frac{2F_0}{1-\Lambda},$$

and check $\Lambda < 1$, $F_0 + \Lambda\varrho \le \varrho$ and $\varrho r^{N_T+1}/\mathsf{w}_j \le b_j$; then $M_j = M_j^0 + b_j$.

Proof. The remainder $R = \mathsf{y} - p$ solves $R = T(R)$, $T(R)(\tau) = \int_0^\tau [d(\tau') + \mathcal{F}(c'+\tau', p+R) - \mathcal{F}(c'+\tau', p)]\,\mathrm{d}\tau'$ along the segment $[0,\tau]$. On the Banach space of R analytic in the open disc and continuous on its closure, with norm $\max_i \mathsf{w}_i \sup_{|\tau|\le r}|R_i(\tau)|/|\tau|^{N_T+1}$, every R with $\|R\| \le \varrho$ has $|R_j| \le \varrho r^{N_T+1}/\mathsf{w}_j \le b_j$, so $p+R$ stays in the convex box B, where the mean-value inequality gives $|\mathcal{F}_i(p+R) - \mathcal{F}_i(p+R')| \le \sum_j L_{ij}|R_j - R'_j|$. Integrating $|\tau'|^{N_T}$ and $|\tau'|^{N_T+1}$ from 0 to $|\tau|$ gives $\|T(0)\| \le F_0$ and $\|T(R) - T(R')\| \le \Lambda\|R - R'\|$, so T maps the ball into itself and contracts; its fixed point gives $\mathsf{y} = p + R$, analytic in the disc, with $\mathsf{y}(c') = \mathsf{y}_0$. □

Lemma G (gluing). *Let y^+ be analytic on an open set G_0. Let $T_1, \dots, T_K$ be open sets, and let each f_m solve the system and be analytic on an open disc $D_m \supset T_m$. Suppose that for each m the set $M_m = T_m \cap (G_0 \cup T_1 \cup \cdots \cup T_{m-1})$ is connected and contains a point q_m with $f_m(q_m) = f(q_m)$, where f is y^+ if $q_m \in G_0$ and otherwise the f_l of an earlier tile $T_l \ni q_m$. Then y^+ extends analytically to $G_0 \cup T_1 \cup \cdots \cup T_K$ and equals f_m on T_m.*

Proof. By induction on m: f_m and the function built so far both solve the system on the connected open set M_m and agree at q_m, so they agree near q_m (the system is regular for $u \ne 0$) and hence on M_m by the identity theorem. □

The covering. Row j has a centre radius ρ_j, a disc radius r_j, a Taylor order N_j and a band $a_j < |u| < b_j$ with $a_j = \rho_j - 0.6r_j$, $b_j = \rho_j + 0.6r_j$; its $K_j + 1$ centres $c_{jk} = \rho_j e^{\mathrm{i}\varphi_{jk}}$, $\varphi_{jk} = -\frac{3\pi}{4}(1 - t_{jk})$, run from the start ray ($t_{j0} = 0$) to the real axis ($t_{jK_j} = 1$), and tile (j,k) is the open polar rectangle $a_j < |u| < b_j$, $|\arg u - \varphi_{jk}| < 0.6 \cdot (3\pi/4)/K_j$:

rows	ρ_j (in u)	$\lvert\xi\rvert$	r_j	N_j	discs	max Λ
0–6	262, 235.36, 208.72, 182.08, 155.44, 128.8, 102.16	0.0618–0.0989	24	70	25, 23, 20, 18, 16, 13, 11	0.191
7–8	79.6, 61.84	0.1121–0.1272	16	60	13, 10	0.164
9–11	47.14, 36.04, 24.94	0.1456–0.2002	10	40	12, 10, 7	0.439

The values at the centres come from the start box at $s_0 = 262\,e^{-3\mathrm{i}\pi/4}$ (the sector version of Lemma 7.2 over X, radius 2.9×10^{-8}), one QR-Lohner run inward along the start ray (order 12, with ω and A as constant states, 452 steps in 11 legs) stopping at each $s_j = \rho_j e^{-3\mathrm{i}\pi/4}$, and one Lohner run along each arc $|u| = \rho_j$ stopping at each centre; each box contains the continuation of y^+ along the ray and the arc, for every $\mathbf{x} \in X$. Lemma D is applied to each box. With $G_0 = \{|u| > 257,\ -3\pi/4 - 0.4 < \arg u < 0\}$, on which y^+ is analytic by Lemma $6\mathrm{b}_0$ and the sector lemma ($|u| > 257$ is $|\xi| < 0.06238$), the tiles are taken row by row from the outside, each row from the ray to the real axis, with agreement points at the midpoints between consecutive centres on each arc (each half-arc lies in its own disc), at ray points between consecutive rows, and at $s_0 \in G_0$. A separate check (`lemma6b1_cover.py check`) verifies in ball arithmetic, for the exact numbers of the construction: (C1) all 178 discs are certified; (C2) every closed tile lies in its open disc; (C3) the half-arcs lie in their discs; (C4) consecutive tiles of a row overlap and non-consecutive ones do not, and the rows reach past both edge rays; (C5) consecutive rows share a ray point inside both discs; (C6) $s_0 \in G_0$; (C7) every meeting set M_m is connected, as a union of polar rectangles linked by proven-nonempty overlaps, and contains its agreement point (566 rectangles, 450 links); (C8) the bands cover $25 \le |u| \le 260$.

Lemma $6\mathrm{b}_1$. *For every $\mathbf{x} \in X$, y^+ extends analytically to an open set containing the annular sector $A_1 = \{25 \le |u| \le 260,\ -3\pi/4 \le \arg u \le 0\}$, that is $0.0620 \le |\xi| \le 0.2$, $0 \le \arg\xi \le 3\pi/8$, and a neighbourhood of it; on each tile, y^+ is enclosed by the Taylor polynomial and the bound M of its disc.*

Proof. Lemma D for each disc (C1), then Lemma G with (C2)–(C7), then (C8). □

All 178 certificates pass, with contraction constants below 0.2 except on the innermost row (0.439), whose start-ray disc comes within 5.7 of u^*; the runs total 6073 Lohner steps. Three tests check that the construction can fail: computing the defect with the fast term of the wrong sign certifies none of the 7 discs of row 11; leaving tile $(5, 6)$ out makes the covering check fail in three ways; and, as a negative control, discs centred at $14\,e^{-3\mathrm{i}\pi/4}$, at distance 5.12 from the branch point u^*, are certified for radii 2 and 3 and not for 4, 4.5, 5.5, 6 or 8, so the disc lemma does not pass on a disc that contains a singularity. Not tested: a system that is wrong in the same way in the Taylor recursion and in the defect; the identity of the system is guarded by self-tests against the other implementations of (7.1).

7.9. **The theorem.**

Theorem A($M = 1$). *Assume Braaksma's 1-summability theorem for the existence of the lateral solutions $\mathsf{y}^{\pm}$ and their asymptotics (Definition 7.1; the one fact taken from the literature). Let $\mathbf{x} = (A, \omega/\omega_\infty)$, $\omega_\infty = 2\pi/1.64389375$, and $X = (2.5421480475, 0.7769940136) + [-10^{-9}, 10^{-9}]^2$. Then the $M = 1$ selection condition has exactly one zero in X, it is simple, and it lies at*

$$A \in [2.54214804761338,\ 2.54214804761975], \qquad \omega/\omega_\infty \in [0.776994013607438,\ 0.776994013612010]$$

($\omega \approx 2.96977670855$). Precisely:

- *there is exactly one $\mathbf{x}^* \in X$ with $\mathsf{y}^+ \equiv \mathsf{y}^-$, equivalently with $\Sigma(\mathbf{x}^*) = 0$, equivalently with $J(0.3; \mathbf{x}^*) = 0$;*
- *$\mathbf{x}^*$ lies in the box above;*
- *the real 2×2 Jacobians of Σ and of $\mathbf{x} \mapsto J(0.3; \mathbf{x})$ are non-singular at $\mathbf{x}^*$.*

Proof. (1) $J^{\mathrm{path}} = J$. Lemma $6\mathrm{b}_0$ gives analyticity of y^+ on the part $|\xi| \le 0.0624$ of $\Omega = \{0 < |\xi| \le 0.2,\ 0 \le \arg\xi \le 3\pi/8\}$, Lemma $6\mathrm{b}_1$ on a neighbourhood of the annular sector, and the two overlap in G_0, where both are y^+. So y^+ is analytic on Ω, the region between the lower path and the real axis; the continuation along the lower path equals the continuation along the real axis, and $J^{\mathrm{path}}(0.3; \mathbf{x}) = J(0.3; \mathbf{x})$ for every $\mathbf{x} \in X$. (2) *One simple zero.* By the Krawczyk test on the lower path, $J^{\mathrm{path}}(0.3; \cdot)$ has exactly one zero in X; it is simple and lies in the box (7.6). (3) *It is the selection condition.* By (1), $\mathsf{y}^+_{\mathbb{R}}(0.3; \mathbf{x})$ is the path value, so it lies in the state set B_1 of the box run; that is the hypothesis of Lemma 6a, which gives $J(0.3; \mathbf{x}) = 0 \Leftrightarrow \mathsf{y}^+ \equiv \mathsf{y}^- \Leftrightarrow \Sigma(\mathbf{x}) = 0$. (4) *Simplicity.* At $\mathbf{x}^*$, $\psi \equiv 0$, so $\varphi_j = \partial\psi/\partial x_j$ solves the variational equation along $\mathsf{y}^+ = \mathsf{y}^-$, with finite weighted norm (each $\psi(\cdot; \mathbf{x})$ obeys the flat estimate). The argument of Lemma 6a applies to φ_j verbatim, since its coefficients lie in the same hulls. So the real-linear map $\mathcal{L} : \partial\Sigma \mapsto \partial J(0.3)$ is injective, $DJ(0.3; \mathbf{x}^*) = \mathcal{L} \circ D\Sigma(\mathbf{x}^*)$, and $D\Sigma(\mathbf{x}^*)$ is non-singular. □

7.10. **Independent checks.**

Code review. An independent review (`proof/A_review.md`) checked that every conversion out of ball arithmetic is a choice, a report, or outside the certificate path, and reproduced every input of the Krawczyk test by separate methods: the start values to 5×10^{-35} (70-digit integration), the point run to 10^{-31} (30-digit mpmath), one Lohner step on each leg against a 50-digit solution (with two negative controls), the vanishing orders exactly in $\mathrm{GF}(p)$, the kernel bounds by quadrature, and $K(X)$ with separate code. It found no error in any enclosure. The review covered steps (i)–(iv) on the ray–arc path. Lemmas S, $6\mathrm{b}_0$, 6a and $6\mathrm{b}_1$ are outside its scope: they reuse the reviewed integrator and majorant code, but their new parts have not been reviewed independently.

A second implementation with CAPD. Steps (ii)–(iv) were redone in C++ with the CAPD library [11], version 6.1.0, sharing no integration or test code with the Python proof. The vector field is written from (7.1) and Taylor coefficients and Jacobians come from CAPD's automatic differentiation [47]; the point run uses 128-bit MPFR intervals with a C^0 doubleton Lohner set at Taylor order 40 (205 steps), and the box runs use double-precision intervals with CAPD's C^1 Lohner set at order 20, started from a mean-value form of the step-(i) data. The Krawczyk test is written anew and run twice, on $g = \kappa_0 J$ and on $(\mathrm{Re}\, J, \mathrm{Im}\, J)$, which needs no normalization at all. On the ray–arc path at radius 0.3 it passes for $h = 10^{-9}$ (margins 8.72×10^{-10}, 9.83×10^{-10}) and for $h = 10^{-8}$ (margins 8.73×10^{-9}, 9.30×10^{-9}), and fails at 10^{-7}, where the radii of $J(X)$ reach the tolerance; its enclosure of the zero, $A \in [2.54214804761512, 2.54214804761800]$, $\omega/\omega_\infty \in [0.77699401360886, 0.77699401361058]$, lies inside that of the Python run on the same path and is eight to ten times narrower. On the lower path it passes for $h = 10^{-9}$ (margins 8.32×10^{-10}, 9.51×10^{-10}) and $h = 10^{-8}$ (margins 4.77×10^{-9}, 6.12×10^{-9}), and its zero box, $A \in [2.54214804760992, 2.54214804762320]$, $\omega/\omega_\infty \in [0.77699401360470, 0.77699401361475]$, contains the box (7.6). The two enclosures of $J(\hat{\mathbf{x}})$ intersect, with midpoints differing by less than 10^{-28}. Five mutation controls (C1, C2, C5, C6, and $J(\hat{\mathbf{x}})$ widened by $10^3 h$) give their expected verdicts at both h.

What is and is not independent. Shared between the two implementations are the equations (7.1), the step-(i) lemma and its code (whose exported start data were checked against an independent 60-digit evaluation of the formal series), and the constant κ_0, which only the κ_0-scaled CAPD test uses. Lemmas S, $6b_0$ and 6a use the Python integrator and majorant code only, and the disc covering of Lemma $6b_1$ (Lemmas D and G and the checks C1–C8) has no second implementation yet; re-running the discs in CAPD would be the natural check.

7.11. **Three harmonics.** The same method certifies the three-harmonic truncation of Section 9.2 (`proof/A_M.md`, `proof/A_lemmas_M.md`). A point certificate (validated integration along the ray $\arg\xi = 5\pi/16$ from 0.0741, the arc $|\xi| = 0.2$ and the real axis to $\xi_1 = 0.3$, then a Krawczyk test on the four real D-jump conditions) shows that they have exactly one zero, with a non-singular Jacobian, in a parallelepiped of outer half-width 10^{-10}, at $A \in [2.72647588563032, 2.72647588563144]$, $\omega/\omega_\infty \in [1.01560309485743, 1.01560309485804]$, $B_3 \in [0.144419478469924, 0.144419478470230]$ and $C_3 \in [0.0110874258881079, 0.0110874258881255]$. Lemmas S, $6b_0$, 6a and $6b_1$ carry over to M harmonics with the same proofs, and at $M = 3$ they are certified on boxes that contain the certificate's (6a with $\rho_{6a} = 0.041$, $6b_1$ by 183 discs in a C^1 mean-value form). So, assuming Braaksma's theorem, the three-harmonic selection condition $S_1 = S_3 = 0$ has exactly one zero in that box, and it is simple. The zero is 1.6% above ω_∞ in frequency, at the measured amplitude. An independent review (`proof/A_M_review.md`) re-derived the Krawczyk inclusion in exact rational intervals, checked single integration steps and the start values against high-precision references, and confirmed the hypotheses of the assembly; the majorant code of lemmas S and $6b_0$ and the disc certificates of $6b_1$ are not yet reviewed.

7.12. **What is not claimed.** Theorem A($M = 1$) is a statement about the one-harmonic truncation ($M = 1$), not about the physical problem: its zero is 22% below ω_∞ in frequency, and it does not survive the addition of harmonic 3. Nothing is said about parameters outside X. The disc covering has no second implementation. At $M = 5$ the higher harmonics leak into the lower conditions with weights up to 10^{17}, so a certificate there must be posed on the Schur-corrected conditions, with multiprecision Lohner integration.

8. The untruncated multiplier as an entire function

8.1. **The amplitude expansion.** Truncating harmonics is not a controlled approximation (Section 4), but there is an expansion in which every harmonic is kept exactly. Write

$$F = A\sin\theta + \sum_{\nu\geq 3} f_\nu(A)\, e^{\mathrm{i}\nu\theta} + \text{c.c.}$$

Harmonic μ first appears at order $A^{|\mu|}$, so keeping $|\mu| \leq K$ at each order in A is exact. At each order the inner equations are linear in the new unknowns, with forcing from lower orders; the late orders in j of each harmonic are normalized by their exact linear form $\Gamma(\alpha + j - \delta/2)\Gamma(\beta + j - \delta/2)$, $\delta = \langle F^2\rangle$, de-mixed by $e^{-\Pi}$, and fitted with every neighbour family ν', $|\nu'| \leq k$, present (they enter through the label θ_0 at relative order $1/j$). Tuning f_ν order by order so that $S_\nu = 0$ for all $|\nu| \geq 3$, the harmonic-1 multiplier becomes a power series in A^2,

$$S_1/S_1^{(1)} = 1 + \sigma_1 A^2 + \sigma_2 A^4 + \dots, \tag{8.1}$$

with $S_1^{(1)}$ the linear multiplier. The coefficients are computed in ball arithmetic at 640 bits from late-order fits; at $\omega = \omega_\infty$ two fit windows, $J = 750$ and 1000 terms, agree to about 10^{-19}:

$$\sigma_1 = 0.74358457992 + 0.52447485276\mathrm{i}, \quad \sigma_2 = 0.11879383784 + 0.39807457035\mathrm{i},$$
$$\sigma_3 = -0.05397185243 + 0.11658742432\mathrm{i}, \quad \sigma_4 = -0.02886475753 + 0.01077835879\mathrm{i},$$
$$\sigma_5 = -0.00555413676 - 0.00280595708\mathrm{i},$$
$$\sigma_6 = (-2.7853745790 - 10.776628405\mathrm{i}) \times 10^{-4}, \quad \sigma_7 = (1.0223846436 - 1.4905053530\mathrm{i}) \times 10^{-4};$$

σ_1–σ_{12} are available at $\omega/\omega_\infty = 0.99$, 1.00, 1.01 (`solver/inner_amp/`). A fit is accurate but not an enclosure; Lemma 8.1 below encloses σ_1 rigorously. Summing (8.1) directly at the zero is hopeless: the terms $|\sigma_k| A_0^{2k}$ grow to about 240 at $k = 9$ and are still 160 at $k = 12$, and would fall below 10^{-6} only near $k \approx 37$.

8.2. **Exponential type.** The coefficients follow a ratio law, $\sigma_k/\sigma_{k+1} \approx (k + \beta_\star)/\kappa_\star$, with constant increments $1/\kappa_\star \approx 0.462 - 0.384\mathrm{i}$ for $k \geq 8$ (`solver/proof/entire_cert.py`):

ω/ω_∞	$\kappa_\star$	type $\lvert\kappa_\star\rvert$	$\beta_\star$
0.99	$1.271 + 1.081\mathrm{i}$	1.668	$4.10 - 1.56\mathrm{i}$
1.00	$1.279 + 1.081\mathrm{i}$	1.675	$4.16 - 1.60\mathrm{i}$
1.01	$1.289 + 1.080\mathrm{i}$	1.682	$4.23 - 1.64\mathrm{i}$

So $|\sigma_k| \approx C|\kappa_\star|^k/\Gamma(k+\beta_\star)$: numerically, S_1 is an entire function of $\mathcal{A} = A^2$ of order 1 and exponential type about 1.67. At ω_∞, $|\sigma_k|\,\Gamma(k+4.16)/1.675^k = 27.3$, 26.8, 26.3, 25.9, 25.6, 25.5, 25.3, 25.1, 24.9 for $k = 4$–12, which suggests the tail bound $|\sigma_k| \le C\,a_\star^k/\Gamma(k+b_\star)$ with $a_\star = 1.675$, $b_\star = 4.16$, $C = 27.3$; it is calibrated on $k \le 12$, not proved. The Borel transform $B(t) = \sum_k k!\,\sigma_k\,t^{-k-1}$ has, by Padé approximation, a continuous singular support rather than isolated exponents: an arc from about $0.75 + 0.16\mathrm{i}$ to $\kappa_\star$, with a density vanishing at $\kappa_\star$ like $(\kappa_\star - \zeta)^{\beta_\star - 1}$. So S_1 behaves like a Laplace integral of a density over that arc.

Factoring out the arc's centre removes the cancellation. With $G = e^{-\bar\kappa\mathcal{A}}S_1$ (same zeros), $\bar\kappa$ chosen per truncation order K to minimize the last terms at the zero, the terms of G at the zero are $O(1)$ and fall to 3.6×10^{-4} by $k = 12$:

K	$\bar\kappa$	last terms at $\mathcal{A}_0$	$A_0(\omega_\infty)$	real zero at ω/ω_∞
9	$0.944 + 0.503\mathrm{i}$	8.1×10^{-3}	$2.71494 - 0.00284\mathrm{i}$	0.99767
10	$1.017 + 0.464\mathrm{i}$	2.9×10^{-3}	$2.72235 + 0.00054\mathrm{i}$	1.00035
11	$0.951 + 0.507\mathrm{i}$	1.0×10^{-3}	$2.71986 - 0.00019\mathrm{i}$	0.99987
12	$0.893 + 0.550\mathrm{i}$	3.6×10^{-4}	$2.71943 - 0.00017\mathrm{i}$	0.99988

The plain Taylor truncation is useless at these orders (its zero drifts from $A = 1.99$ at $K = 6$ to 2.53 at $K = 12$). The other zeros of G_K lie at $|\mathcal{A}| > 10$ and move outward with K, so the zero $\mathcal{A}_0 = A_0^2 \approx 7.40$ is isolated.

8.3. **What a certificate would need.** A Krawczyk test for $(\mathrm{Re}\,G, \mathrm{Im}\,G)$ in $(A, \omega/\omega_\infty)$, where the Jacobian has singular values 0.53 and 0.26, needs the total error below about 2.6×10^{-3} for a box of radius 0.01 and 2.6×10^{-4} for 10^{-3} (Rouché's theorem on $|\mathcal{A} - \mathcal{A}_0| = 0.01$ at fixed ω needs 5.7×10^{-4}). A type bound calibrated at $k = 12$ puts the tail at about 2×10^{-4}, so $K = 12$ would certify a box of radius 0.01 and $K \approx 14$–15 one of radius 10^{-3}, *if* the coefficients and the tail were rigorous. That needs: (1) rigorous σ_k for $k \le 14$, each a Stokes constant of a linear inhomogeneous system, enclosed by validated lateral integration or by an exact pairing formula (standard but laborious); (2) a tail bound, which is the hard part: a majorant on the recursion gives only geometric bounds, so the bound must come from control of the inner problem at complex amplitude $|A| \approx 4$–4.5 (Cauchy estimates on a circle) or from a Laplace representation proving the exponential type; and (3) uniform control in ω between the scanned values. A plausible mechanism for the type is that, to leading order, $\langle F^2\rangle = \mathcal{A}/2$, so the late-order normalization $\xi^{2\langle F^2\rangle}$ is $\xi^{\mathcal{A}}$, integrated against an $\mathcal{A}$-independent kernel along a bounded complex path; writing order $2k+1$ as a k-fold Borel convolution over a simplex, whose volume supplies the $1/k!$, is the proposed route to a proof. This replaces the harmonic tail (gap (b)) by an amplitude tail that is much better behaved. A certified box around 0.9999 of radius 10^{-3} would contain ω_∞, but containing it is not proving equality: that needs the selection theorem.

8.4. **A rigorous first coefficient.** For σ_1 the late-order fit can be replaced by an exact formula (`solver/proof/exact_sigma.py`). At order 3 the amplitude recursion for harmonic +1 is a first-order linear recursion with coefficients $a_j = (j+\alpha)(j+\beta)/(j+1)$ and an explicit forcing built from order-1 sequences and the order-2 lapse coefficients; it solves in Darboux form, and the multiplier sequence R_j (normalized by the exact linear form, which times $T_j = \prod_{m<j} a_m$ is exactly $1/(\Gamma(\alpha)\Gamma(\beta))$, and de-mixed) becomes an explicit telescoping sum plus digamma terms, so that $S^{(3)} = R_J + \sum_{j\ge J}\Delta R_j$ exactly and $\sigma_1 = S^{(3)}/S^{(1)}$. Three facts make the tail computable: a *cancellation lemma*, by which the $O(1/j)$ parts of ΔR_j cancel exactly in two pairs, leaving explicit rational functions of j (degree 3 over 5, one of them multiplied by a unimodular rotating factor) plus a remainder r_j; a *remainder bound* $\sup_{j\ge J} j^3|r_j| \le C_3/|\Gamma(\alpha)\Gamma(\beta)|$, with $C_3 = 110.3$ at ω_∞ and $J = 800$, from the log-convexity of $|T_l|$ for $l \ge 1$; and *closed-form tail sums* of the rational parts, by partial fractions in digamma and Hurwitz zeta functions and two summations by parts.

Lemma 8.1 (σ_1; computer-assisted)**.** *At $\omega/\omega_\infty = 0.99$, 1.00 and 1.01, as point values of ω ($\omega_\infty = 2\pi/1.64389375$), with $J = 1600$, the first coefficient σ_1 of (8.1) lies in the boxes (componentwise, centre $\pm$ half-width)*

ω/ω_∞	$\mathrm{Re}\,\sigma_1$	$\mathrm{Im}\,\sigma_1$	*half-width*	*fitted* σ_1 *minus centre*
0.99	*0.736457587*	*0.528917680*	3.46×10^{-5}	2.4×10^{-6}
1.00	*0.743582522*	*0.524475851*	3.54×10^{-5}	2.3×10^{-6}
1.01	*0.750661133*	*0.520061240*	3.62×10^{-5}	2.1×10^{-6}

which contain the fitted values.

The enclosures hold all rounding errors, the j^{-3} remainder and the summation-by-parts remainder; they are about 60 times inside the accuracy a 10^{-3} certificate needs for σ_1. The only inputs are the definition of R_j (the de-mixing constant E_2, the normalization h_2 and the recursion), taken from the amplitude recursion as the definition of the multiplier. The code has not been reviewed independently: its recursion is an implementation separate from the amplitude code, whose R_j it reproduces to 10^{-39} for $j \le 400$, but no second agent has checked the certificate. Not yet done: uniformity in ω over a box (with ω as a ball the enclosure is useless already at radius 10^{-7}, because rectangular complex balls wrap through J multiplications), and σ_2–σ_6, which telescope the same way with increments $O(\log^k j/j^2)$ and need the closed-form endpoint sums from σ_2 on; this is mechanical but long.

9. Numerical evidence for Conjecture 1

Nothing in this section is a proof. It records what the selection rule gives when it is evaluated numerically, and how that compares with the critical solutions of [1].

9.1. **Two implementations.** The inner multipliers are computed by two implementations that share no code: *lateral solutions* (integrated on either side of the Stokes line and differenced) and *Borel–Darboux* (late orders fitted in ball arithmetic with the exponents of Section 4). Without gravity both reproduce the exact Stokes constant, to 10^{-11} and 6×10^{-27}.

9.2. **The truncation sequence.** *One harmonic.* With $F = A\sin\theta$ the multiplier has an isolated zero at $A = 2.542148$, $\omega = 2.969777$ ($\Delta = 2.11571$), found by both methods (lateral, the same at three matching radii; Darboux, with Newton's method converging quadratically to $|S| = 3\times10^{-14}$ at 220 orders, and the only clean zero in $0.5 \le A \le 3$, $0.5 \le \omega/\omega_\infty \le 1.5$). This is the zero that Theorem A($M = 1$) certifies. Its amplitude is 6.5% below the measured $F_1 \approx 2.72$ and its frequency 22% below ω_∞; adding harmonic 3 destroys it (the residual becomes 4.6×10^{6}).

Three harmonics. A joint solve of $S_1 = S_3 = 0$ is badly conditioned (singular values 6×10^{9} down to 0.4) and stops on false near-zeros. Solved nested ($S_3 = 0$ for F_3 at fixed (A,ω), then a 2×2 Newton step for $S_1 = 0$), there is one zero, the same from two starts and at three matching radii (moving by less than 10^{-7}): $A = 2.7265$, $\omega = 1.0156\,\omega_\infty$ ($\Delta = 1.6186$), $F_3/F_1 = 0.0530 + 0.0041\mathrm{i}$, $\langle F^2\rangle = 3.727$. Its shape is close to that of the $D = \infty$ solution extrapolated from the data ($A \approx 2.70$–2.72, $|F_3/F_1| \to 0.0549$–0.0552, $\langle F^2\rangle \approx 3.7$ [1]), while its frequency is 1.56% above ω_∞, against 22% below with one harmonic.

Five harmonics. The couplings grow further ($\partial r_3/\partial B_5 \approx 10^{13}$, $\partial r_1/\partial B_5 \approx 10^{17}$). A hierarchical solve ($S_5 = 0$ innermost, then $S_3 = 0$, then $S_1 = 0$ for (A,ω)), with a Schur-complement correction that evaluates r_1 as if the higher harmonics sat exactly on their zeros, gives $A = 2.718579$, $\omega = 0.999228\,\omega_\infty$ ($\Delta = 1.64516$, 0.077% above Δ_∞), with $|r_1|$ down to 8×10^{-7}. At fixed $(A,\omega) = (2.72,\omega_\infty)$ it gives $F_3/F_1 = 0.05486 + 0.00541\mathrm{i}$ and $F_5/F_1 = 0.00309 + 0.00054\mathrm{i}$, against $0.05482 + 0.00543\mathrm{i}$ and $0.00310 + 0.00056\mathrm{i}$ from the amplitude expansion, which shares no code with it; the differences are of the size of the omitted F_7.

The offset in ω thus goes -22%, $+1.56\%$, -0.077% for $M = 1$, 3, 5: it alternates in sign and shrinks fifteen- to twentyfold per added pair of harmonics, and Aitken's extrapolation [3] of the three gives $\omega/\omega_\infty = 1.0003$ (three terms are a thin basis). With the ratio of -0.045 to -0.057 per pair of Section 6.6, the prediction for $M = 7$ is $\omega/\omega_\infty = 0.99997$–$1.00017$. The seven-harmonic zero, computed with a compiled backend at matching radius $\xi_1 = 0.15$ (`proof/B_tail_M7.md`), is $A = 2.71922$, $\omega/\omega_\infty = 1.000022$ ($\pm2\times10^{-5}$, from the scatter of repeated solves; at matching radii 0.14 and 0.16 the solves did not converge well enough to give a zero), inside the prediction. The last ratio, -0.049 per pair, is the decay of the added pair (0.057) times a coupling of 0.85, and the geometric limit of the sequence is $\omega/\omega_\infty = 0.999985$.

9.3. **The shape of the echo.** Tuning the higher harmonics in the amplitude expansion of Section 8.1, at $A = 2.72$ and $\omega = \omega_\infty$, gives $|F_3/F_1| = 0.0551$ and $|F_5/F_1| = 0.0031$ (0.00314 with more digits). The critical solutions at $D = 20$–60, extrapolated in $1/n$, give $|F_3/F_1| \to 0.0552$–0.0549 and $|F_5/F_1| \to 0.0034$–0.0032 [1]. This is a prediction of the higher-harmonic conditions $S_\nu = 0$, $|\nu| \ge 3$, alone, checked against the solutions, and it agrees. The tuned ratios are $f_{\nu+2}/f_\nu \approx qA^2$ with q nearly independent of ν, so $F \approx 2\,\mathrm{Re}\left[F_1 e^{\mathrm{i}\theta}/(1 - qA^2e^{2\mathrm{i}\theta})\right]$,

with singularities at $|\operatorname{Im}\theta| \approx \frac{1}{2}\ln(1/|q|A^2) \approx 1.4$: F_∞ is then not entire, and its harmonics decay at a finite rate. The measured ratios $|F_{\nu+2}/F_\nu|$, $\nu = 1, \dots, 9$, extrapolate to 0.055–0.072, a decay rate of 1.3–1.45 per harmonic, close to the tuned profile's decay of 0.0574 per pair of harmonics (Section 6.6; $|q|A^2 \approx 0.063$ at leading order). The extrapolations, which are not fully converged, are plotted in [1]. This is the evidence behind the value $\sigma_F \approx 1.4$ in hypothesis (H1).

9.4. **The harmonic-1 condition.** At ω_∞ the coefficients τ_k of $\ln S_1$ in A^2 have $\arg\tau_k/\pi = 0.878$, 0.944, 0.978, 0.986, 0.992, 0.995 for $k = 2$–7, approaching 1, as a real zero A_0^2 requires ($\tau_k \to -1/(kA_0^{2k})$; a complex zero would turn them with k). Padé approximants [1/5] to [5/1] of $(\ln S_1)'$ put the zero at $A_0 = 2.728$–2.756 with $\operatorname{Im} A_0$ between -0.013 and -0.027: within about 1% of the amplitude 2.70–2.72 measured on the solutions.

Across ω, two families of estimates of where the zero is exactly real approach ω_∞ from opposite sides (Table 3): the crossings $\arg\tau_k = \pi$, which assume a simple zero, and the ratio estimates, which let the exponent float. The coefficients behave as $-c_{\text{eff}}/(kA_0^{2k})$ with an effective exponent c_{eff} that falls toward 1, so the zero is simple and the crossings from above are the asymptotically correct estimator; their distance from 1 shrinks by a factor of about 0.4–0.6 per order, and Aitken's extrapolations of consecutive triples scatter around 1 at the 10^{-4} level. With twelve coefficients the five central Padé approximants of $(\ln S_1)'$ put the real zero at 0.99987–0.99991 (at ω_∞ they are real to $|\operatorname{Im} A_0| \approx 1.4 \times 10^{-4}$, at $A_0 = 2.7195$–2.7198), and the factored series G of Section 8 gives 0.99988 at $A_0 = 2.7194$. The truncations give 0.999228 ($M = 5$) and 1.000022 ($M = 7$).

TABLE 3. Where the zero of S_1 becomes real, in ω/ω_∞, order by order in the amplitude expansion.

order k	$\arg\tau_k = \pi$ (from above)	τ_{k-1}/τ_k real (from below)	effective exponent c_{eff} at ω_∞
5	1.0083	0.9821	
6	1.0040	0.9803	1.148 − 0.159i
7	1.0021	0.9913	1.177 − 0.095i
8	1.0011	0.9946	1.077 − 0.059i
9	1.0006	0.9961	1.052 − 0.044i
10	1.0003	0.9978	1.021 − 0.026i
11	1.00020	0.99919	1.017 − 0.012i
12	1.00011	0.99915	1.013 − 0.012i

A planted-truth test (in the repository, `notes/PLANTED_SELECTION.md`) ran these estimators, through the same code, on synthetic functions of the same class: Laplace integrals over a Borel arc whose coefficients match the real $\sigma_0, \dots, \sigma_{12}$ to 10^{-5}, with the zero planted at $\omega/\omega_\infty = 0.996$–$1.004$. The pair 0.99915–1.00011 is not a bracket: it misses one family of shapes, and its lower end comes from the ratio estimate, which is 1.3–5.4×10^{-3} low on every plant. The Padé and factored estimates are biased in opposite directions by up to 8×10^{-4}, so their agreement on the real data is not evidence of accuracy. The crossing estimate is the most reliable (errors -0.8 to $+1.9 \times 10^{-4}$); at twelve orders, calibrated on the plants, it put the zero at $\omega^*/\omega_\infty = 0.99992$–$1.00019$, at $A_0 = 2.720$ (solutions: 2.70–2.72), if S_1 has no zeros besides x_0 within $|x| \approx 9$. Twelve coefficients do not exclude them: functions with exactly the real $\sigma_0, \dots, \sigma_{12}$ have their zero anywhere in 0.996–1.002, the far cases hiding a ring of zeros.

With σ_{13} converged (960 bits, relative error 2×10^{-13}), the crossings keep falling (1.000299, 1.000197, 1.000109, 1.000046 at $k = 10$–13) and the factored series gives 0.999995. Every ring completion of $\sigma_0, \dots, \sigma_{12}$ constructed so far mispredicts the thirteenth coefficient of $\ln S_1$ by 10–300%, while the ring-free simple-zero law misses it by 8×10^{-4}. Calibrated on the plants at thirteen orders, the crossing and factored estimates put the zero at 0.99992–0.99996, and the ring-free functions that reproduce $\sigma_0, \dots, \sigma_{13}$ exactly, [6/7] and [5/8] Borel–Padé sums, at 0.99998 and 0.99996.

With σ_{14} converged (relative error 3×10^{-13}; a 1280-bit run agrees in every printed digit), the crossing falls to 1.000026 and the effective exponent reaches 1.000. Every ring completion of $\sigma_0, \dots, \sigma_{13}$ mispredicts the fourteenth coefficient of $\ln S_1$ by 35–390%, while the ring-free Borel–Padé sums of $\sigma_0, \dots, \sigma_{13}$ predicted σ_{14} to 1–6×10^{-7}; those of $\sigma_0, \dots, \sigma_{14}$ put the zero at 0.99998–0.99999. Calibrated on the plants at fourteen orders, the crossing and factored estimates put it at 1.00004–1.00005: the calibrated value moved by more than either range's width, because the planted functions do not follow the real coefficients from one order to the next. Taken over both orders, and read as a value of the echo period,

$$\Delta_\infty = 1.6439 \pm 0.0001 \qquad (1.64381\text{–}1.64403)$$

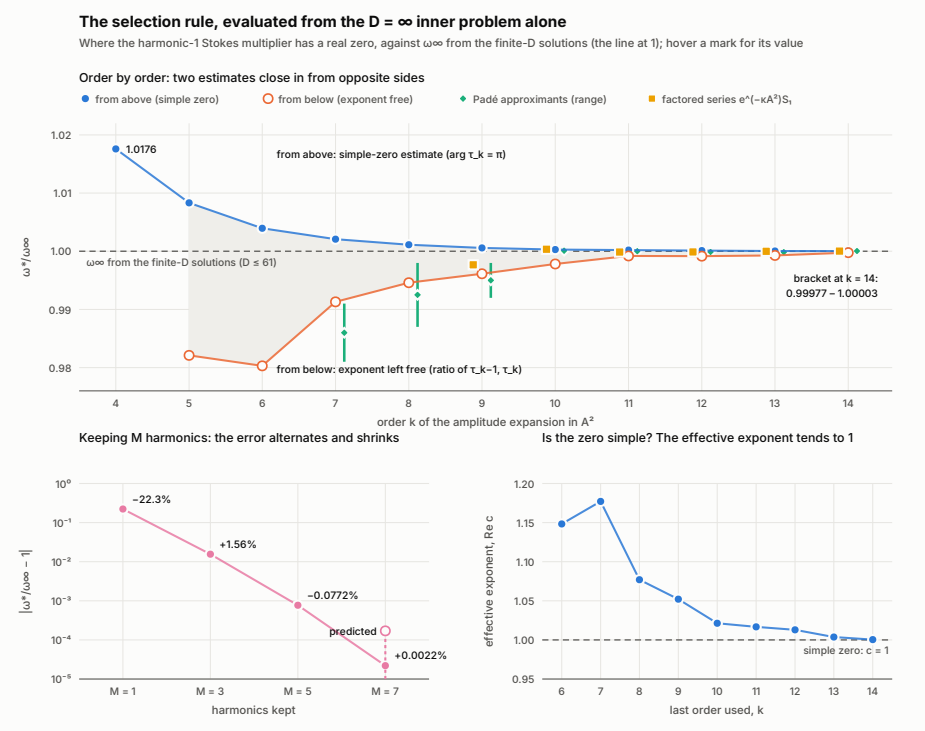


FIGURE 1. The selection rule evaluated from the $D = \infty$ inner problem alone. Top: where the harmonic-1 multiplier has a real zero, order by order in the amplitude expansion; simple-zero estimates ($\arg \tau_k = \pi$) from above, floating-exponent estimates from below, Padé ranges and the factored series inside the shaded bracket, which narrows onto ω_∞ from the finite-D solutions (dashed). Bottom left: the truncated route, $|\omega^*/\omega_\infty - 1|$ for $M = 1, 3, 5, 7$ harmonics (alternating in sign), with the $M = 7$ range predicted from $M \le 5$ (dashed). Bottom right: the effective exponent of the zero tends to 1.

from the $D = \infty$ inner equations alone, with no input from the finite-D solutions, against 1.643893746 from them [1], which lies inside the range. The range holds only if S_1 has no zeros besides x_0 within $|x| \approx 12$; a ring of fifteen or more zeros is still possible, and σ_{15} is its test.

9.5. **What the evidence does not show.** The ± 0.0001 above is the range of calibrated estimators, not an error bound, and it rests on the assumption about the other zeros of S_1. The coefficients σ_k come from late-order fits; two fit windows agree to about 10^{-19} for σ_1–σ_{12} (σ_{13} to 2×10^{-13} relative; σ_{14} was fitted with one window, to a plateau step of 3×10^{-13} relative), and the estimates need them to about 10^{-10}, but only σ_1 is enclosed rigorously (Lemma 8.1). At the zero the series (8.1) is far from summed, and every estimator (crossings, ratios, Padé approximants, the factored series) is a resummation whose convergence is observed, not proved; the arc-shaped Borel singularity of Section 8 is a known source of slow convergence. Settling the other zeros needs more orders, starting with σ_{14}. Finally, agreement of the numbers is evidence that the selection rule is the right one, not a proof of Conjecture 1.

9.6. **Other matter.** The same rule passes tests on continuously self-similar critical solutions, where no harmonic truncation enters: for the axion–dilaton and the free complex scalar the zero of the $D = \infty$ Stokes multiplier agrees with the frequency extrapolated from finite D to 1.6×10^{-4} and 1.3×10^{-4} (relative), and for a σ-model it agrees with the extrapolated ground state within the extrapolation's error [1].

10. OUTLOOK

The large-D limit of Choptuik's critical solution is a selection problem of the Saffman–Taylor type, with a vector of harmonics in place of a scalar parameter. The linear version of the whole chain (outer late orders, inner Stokes constant, light-cone coefficient) is proved. What would turn the selection rule into a theorem, in order of difficulty:

(1) *Theorem B(M) for* $M = 1$: the missing items of Section 6.5, chiefly a certified error bound in Lemma B5(2) (or a matching domain of n-independent size, `proof/B5_inner.md`, §5.4) and independent reviews of the certificate code; then the same for each fixed M. A proof of (H-σ) would identify the centre-regular solution with the Borel sum of the outer series, but the theorem does not need it.
(2) *Theorem A(*$M = 5$*)*, on the Schur-corrected conditions; an independent review of the $M = 3$ certificate (Section 7.11); and a second implementation of the disc covering.
(3) *The untruncated zero*: the harmonic-tail lemma of Section 6.6 with $M = 7$ explicit, or rigorous σ_k, $k \le 14$, with a proof of the exponential type of S_1 (Section 8).
(4) *Uniformity in the harmonic index* (gap (b)) for the selection theorem itself, in a space of functions analytic in a θ-strip.
(5) *Identification*: that the computed branch of critical solutions is the branch of the selection theorem, for instance by a computer-assisted existence proof in the manner of Reiterer and Trubowitz [43] at one large D.

Together these would derive $\Delta_\infty = 2\pi/\omega^*$ from the $D = \infty$ equations. Numerically the rule already gives Δ_∞ to about 10^{-4}; more orders, or $M = 9$, would sharpen that, and whether Δ_∞ has a closed form is open. For the critical exponent, open are the uniqueness of the unstable mode at $D = \infty$ and the coefficient of $1/n$ in λ. A centrifugal term of order n/r^2 (scalar multiplets with angular momentum, wave maps, Yang–Mills) becomes an explicit $1/s$ in the inner equations and removes the symmetry of Proposition 5. Other matter models test this, and the companion paper reports the tests [1]: for the axion–dilaton, whose inner problem keeps the symmetry, $\gamma \to 1/2$; for the $\ell = 1$ scalar multiplet the computations to $D = 9$ do not decide γ_∞; and the continuously self-similar branch of the σ-model ends at a fold near $D = 8.54$, before large D is reached.

How this work was done

N. M. chose the problem, set the direction at each step and ran the long computations on his own machine. The code, the computations, the derivations, the proofs and the text, including this paper, were produced by Claude, Anthropic's AI model, working as a set of coordinated agents. N. M. has not checked the mathematics himself, and no human expert has reviewed the work.

Checks of the results. Propositions 1–3 and the inner equations were re-derived by separate agents that had not written them, with sympy where possible; the inner Stokes multiplier was computed by two implementations that share no code; the computer-assisted proof of Theorem A($M = 1$) was reviewed by an independent agent, which re-derived every input of its final test and found no error, and steps (ii)–(iv) were re-implemented in C++ with CAPD, sharing no integration or test code; and deliberately broken versions of the final test, of Lemma S, of Lemma 6a and of the disc covering are caught, with the one documented exception in Lemma 6a. Not independently re-checked: Lemmas S, $6b_0$, 6a and $6b_1$ of the proof of Theorem A($M = 1$) (Section 7 lists what is and is not independent), and the enclosure of σ_1 (Lemma 8.1).

Checks of the text. A separate agent wrote a simulated referee report on the combined draft from which this paper and its companion [1] were split. The citations of that draft were checked against publisher or arXiv records, and its novelty claims against the closest prior work, read rather than recalled; the references added since were checked by web search against arXiv, publisher or catalogue records, and the fields that could not be confirmed are marked. After the split, the numbers in the two papers were cross-checked against the repository (every number of three or more significant digits was found in a repository file), and the two papers against each other. Corrections are welcome.

Data and code. All data, code and derivation scripts are at https://github.com/nmoinvaz/choptuik-large-d; the certificate is reproduced by the commands in `proof/A_certify.md`, and the statements of the proof roadmap are in `PROOF.md` and `proof/`.

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